



Electronic Design

An open-source, single-channel DC power supply (Ewigvolt)

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May 7, 2026

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I General description of the design

II Problem Identification

II-A Project Motivation

Since the early development of electronics, the use of power supplies has been essential to operate any electronic circuit. Although there are multiple solutions that provide fixed voltage outputs, laboratory environments require more versatile equipment, particularly power supplies with adjustable voltage and current, operating under constant voltage (CV) and constant current (CC) modes, with adequate stability, low ripple, and ease of use.

Currently, a wide range of DC laboratory power supplies is available on the market. Entry-level switching supplies with digital interfaces, such as low-cost adjustable units, can typically be found in the range of 70–80 USD [1] (shipping not include). These devices generally fulfill the basic requirement of powering electronic systems; however, their performance and usability are often limited.

A key limitation in modern commercial power supplies is the user interface. While many devices are labeled as “digital,” most of them do not provide direct numeric input through a physical keypad. Instead, they rely on rotary encoders and incremental adjustment mechanisms, where the user must select and modify individual digits sequentially. For instance, commonly available low-cost supplies explicitly implement “precision encoder knobs” to adjust values digit by digit rather than through direct numerical entry [1].

This interaction model introduces inefficiencies when fine resolution is required, particularly in laboratory environments where voltage and current must be set quickly and accurately at the millivolt or milliampere level. As a result, even though digital control is present, the lack of an efficient numeric input interface limits usability and precision in practice.

In addition to interface limitations, other issues persist, including restricted access to detailed design documentation, limited insight into internal control strategies, and reduced educational value. Most commercial solutions do not provide sufficient information to analyze aspects such as regulation loops, sensing techniques, or electromagnetic interference (EMI) considerations.

This situation reveals a gap between affordability, usability, and accessibility of design knowledge. In particular, there is a lack of openly documented power supply designs that simultaneously provide:

- Adjustable operation in both CV and CC modes
- Adequate electrical performance (low ripple and stable regulation)
- Efficient and precise user input mechanisms
- Full design transparency and reproducibility

To address this gap, this project proposes the design and implementation of a DC laboratory power supply from the ground up, with fully open and accessible documentation. The system is intended to operate within the following target specifications:

- Output voltage range: 0–30 V
- Output current range: 0–3 A
- Maximum output power: ≥ 90 W
- Voltage resolution: ≤ 10 mV
- Current resolution: ≤ 5 mA
- Maximum Output ripple: ≤ 50 mV
- Load regulation: within 2%

The proposed design emphasizes not only functional performance but also usability improvements, particularly through the implementation of a more efficient input interface, as well as complete transparency in all design stages. This approach enables reproducibility and facilitates its use as both a practical laboratory tool and an educational platform

II-B ODS

The development of Ewigvolt is mainly related to SDG 4: Quality Education and SDG 7: Affordable and Clean Energy. First, by being proposed as an open-source, well-documented, and reproducible DC laboratory power supply, the project supports practical learning in power electronics, control systems, PCB design, prototyping, and validation. It allows students and technical users not only to operate the equipment, but also to understand the design decisions behind its architecture and performance. Second, the project contributes to SDG 7 at a laboratory scale by promoting the design of safer, more efficient, and measurable electrical power systems, considering aspects such as voltage and current regulation, output ripple, efficiency, power factor, and electrical protection. Therefore, the power supply is not only intended as a functional instrument for powering electronic circuits, but also as an educational platform that encourages the responsible, efficient, and accessible use of electrical energy.

III Background and State of the Art

III-A Fundamentals of Power Supplies

Any regulated voltage source relies on design techniques and concepts drawn from multiple areas of electronics, including, but not limited to:

- Power electronics.
- Wideband amplification techniques.
- Operational amplifier applications.
- Control systems.

The full benefits of the engineering behind modern regulated power supplies cannot be realized unless the user first recognizes their inherent versatility and high-performance capabilities, and then understands how to properly apply these features [2].

III-A1 Basic Concepts and Definitions

It is proposed to review certain theoretical and practical concepts, as well as to define practical constants that will be considered throughout the analysis [2].

Ambient Temperature: This is defined as the temperature of the environment surrounding the device. It considers only the site temperature and not the increase caused by the system's own heating. In *Bogotá*, this value is approximately $\approx 15^{\circ}\text{C}$, however, as a common engineering practice, an ambient temperature of 25°C is assumed.

Single-Channel Power Supply: A single-channel design is intended; therefore, multi-output configurations are not considered. Consequently, the following operating modes are outside the scope of this work:

- **Automatic (AUTO) Parallel Operation:** a master-slave configuration in which the outputs of two or more supplies are connected in parallel to increase the available output current.
- **Automatic (AUTO) Series Operation:** a master-slave configuration in which the outputs of two or more supplies are connected in series to increase the available output voltage.
- **Automatic (AUTO) Tracking Operation:** a master-slave configuration in which two or more supplies operate with coordinated outputs, typically maintaining a defined relationship between their voltages.

Drift: Drift refers to the maximum variation in the output of a power supply over a specified period of time (typically 8 hours) after an initial warm-up, while all operating conditions such as load, input voltage, and ambient temperature are kept constant. It includes both slow (periodic) and random deviations, commonly grouped as periodic and random deviations (PARD), within DC range up to 20 Hz.

Inrush Current: Inrush current is the maximum instantaneous input current drawn by a power supply at the moment AC power is first applied.

Load Effect (Load Regulation): Load effect, also known as load regulation, is the change in the steady-state output voltage or current caused by a variation in the load, while all other operating conditions remain constant.

Off-Line Power Supply: An off-line power supply is one whose input rectifier operates directly from the AC mains, without transformer isolation.

Output Impedance of a Power Supply: The output impedance of a power supply is defined as the ratio $\Delta V_{\text{out}}/\Delta I_{\text{out}}$ in response to a change in load, typically as a function of frequency. Ideally, a constant voltage power supply has zero output impedance, while a constant current power supply has infinite output impedance.

Remote Sensing (Remote Error Sensing): Remote sensing is a technique used in constant voltage power supplies to regulate the output voltage directly at the load terminals rather than at the power supply output. This is achieved by adding two low-current sensing leads that measure the voltage at the load as shown on Figure 1.

Remote sensing compensates for voltage drops (IR drops) along the power leads caused by their finite resistance. In a conventional connection, these drops result in a lower voltage at the load than the programmed setpoint. With remote sensing, the power supply

regulates its output based on the voltage measured directly at the load terminals, adjusting its output to maintain the desired voltage at that point. This effectively eliminates errors introduced by wiring resistance, within specified compensation limits.

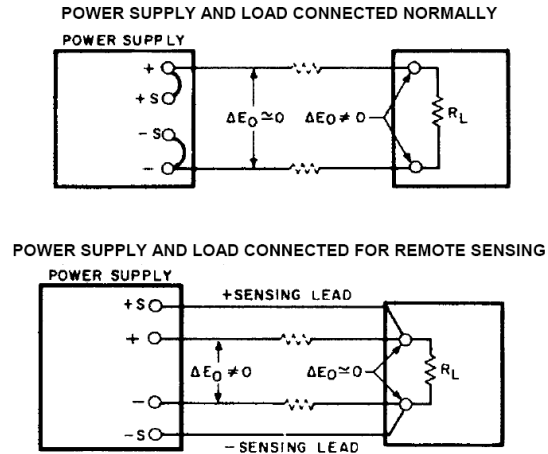


Fig. 1: Remote sensing simplified circuit [2]

This technique is implemented to improve regulation accuracy at the load, particularly in applications involving non-negligible load currents, long or resistive leads, or when high precision (on the order of millivolts) is required. By relocating the feedback point from the supply terminals to the load, remote sensing ensures that the controlled variable corresponds to the actual operating conditions of the load rather than the internal output of the power supply.

Source Effect (Line Regulation): Source effect, also known as line regulation, is the change in the steady-state output voltage (for a CV supply) or current (for a CC supply) caused by a variation in the input (AC line) voltage, while all other conditions remain constant. It is typically evaluated over the full input voltage range, from minimum to maximum line conditions.

There are additional concepts relevant to the proper design stage; however, these are either addressed where their application or evaluation is required, or assumed to be known as part of the reader's prior background.

III-A2 General Regulation Techniques

Modern power supplies are no longer adequately described by a small set of classical regulation categories. Instead, they are more accurately classified according to their power architecture, control strategy, and underlying conversion topology. The modern classification goes by [2], [3]:

Power Architecture:

- **Linear Regulation:** Uses a series pass element operating in the linear region to regulate the output. It provides low noise and excellent regulation at the expense of efficiency.
- **Switch-Mode Power Supply (SMPS):** Employs high-frequency switching converters to achieve high efficiency and power density. Common topologies include buck, boost, flyback, forward, and resonant converters.
- **Hybrid Architecture (Preregulator + Linear):** Combines a switching preregulator with a linear post-regulator. The switching stage provides coarse regulation and efficiency, while the linear stage ensures low ripple and precise output control. This architecture is widely used in high-performance laboratory power supplies.

Control Strategy:

- **Analog Control:** Traditional feedback loops implemented using operational amplifiers and analog circuitry.
- **Digital Control:** Regulation implemented using microcontrollers or digital signal processors (DSPs), enabling programmability, advanced features, and adaptive control.
- **Mixed-Signal Control:** Combines analog feedback loops with digital supervision and configuration, which is common in modern programmable power supplies.

Switching Topologies (for SMPS):

- **Non-Isolated Topologies:** Buck, boost, and buck-boost converters.

- **Isolated Topologies:** Flyback, forward, push-pull, half-bridge, full-bridge, and LLC resonant converters.

Classical Techniques (Legacy Classification): Historically, power supplies were categorized into a small number of regulating techniques:

- **Series (Linear) Regulation**
- **Preregulator + Series Regulator**
- **Switching Regulation**
- **SCR (Thyristor-Based) Regulation**

While these categories remain useful for introductory understanding, they are no longer sufficient to describe modern designs. In particular, SCR-based regulation is largely obsolete in contemporary laboratory power supplies and is mainly restricted to legacy systems or high-power industrial applications. Modern designs predominantly rely on switching or hybrid architectures, often combined with advanced (frequently digital-assisted) control strategies.

III-A3 Performance Metrics: Regulation, Ripple, and Efficiency

Efficiency: Efficiency, expressed as a percentage, is defined as the ratio of the output power of the supply to the active input power. It is typically evaluated at maximum rated output power and under worst-case input voltage conditions, as indicated in Equation 1:

$$\eta = \frac{P_{\text{out}}}{P_{\text{in}}} \times 100\% \quad (1)$$

In practical power supplies, efficiency is limited by conduction and switching losses in semiconductor devices, as well as losses in magnetic components and passive elements. Linear regulators generally exhibit lower efficiency due to continuous power dissipation in the pass element, while switched-mode power supplies (SMPS) achieve significantly higher efficiency by operating devices in switching regimes [2], [4]. Additionally, efficiency varies with load conditions, typically reaching a peak near nominal operating points and decreasing at light load due to fixed switching and control losses [5].

Regulation: Voltage regulation characterizes the ability of a power supply to maintain a constant output voltage under varying operating conditions. It is commonly divided into line regulation and load regulation. Line regulation refers to the variation in output voltage due to changes in input voltage, whereas load regulation describes the variation caused by changes in load current. These parameters are critical in laboratory power supplies, where precise and stable output is required [2]. Mathematically, load regulation can be expressed as:

$$\text{Load Regulation} = \frac{V_{\text{no-load}} - V_{\text{full-load}}}{V_{\text{nominal}}} \times 100\% \quad (2)$$

High-performance designs aim to minimize these variations through tight feedback control and proper compensation techniques.

Ripple and Noise: Ripple and noise represent unwanted AC components superimposed on the DC output voltage. Ripple is primarily associated with switching action and rectification processes, while noise includes high-frequency components generated by fast switching transitions and electromagnetic interference (EMI). These effects are typically quantified as peak-to-peak voltage over a specified bandwidth [2].

In SMPS designs, ripple is influenced by switching frequency, output filtering, and layout parasitics. Increasing the switching frequency can reduce the amplitude of ripple by allowing smaller filter components; however, it also introduces higher switching losses and electromagnetic interference [6]. Consequently, proper filter design and PCB layout are essential to meet performance specifications.

Overall, efficiency, regulation, and ripple constitute the primary performance metrics used to evaluate and compare power supply designs, and they are often interdependent, requiring careful trade-off analysis during the design process.

III-B Power Supply Architectures

III-B1 Quick Analysis of Regulation Circuits

Series Regulation: Series regulation consists of placing a control element in series between the rectifier and the load. This element operates in the linear region and adjusts its effective resistance in order to maintain a constant output voltage. A simplified representation is shown in Fig. 2, where the series element is modeled as a variable resistor controlled by a feedback loop.

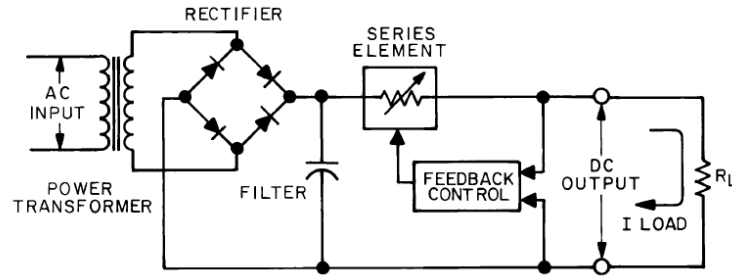


Fig. 2: Series AC-DC converter [2]

In practical implementations, the series element is typically realized using one or more transistors operating in linear mode. The output voltage is continuously monitored and compared with a reference value, generating an error signal that regulates the conduction of the pass element. A representative implementation is shown in Fig. 3:

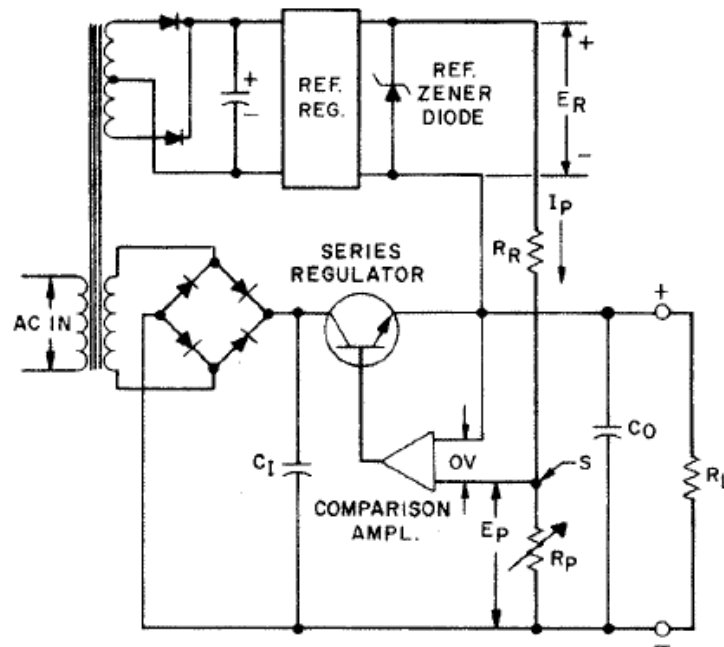


Fig. 3: Example of series regulator implementation [2]

Pros and Cons of Series-Regulated Power Supplies: Series-regulated power supplies provide precise output regulation, fast transient response, and very low ripple and noise. Their simplicity also contributes to high reliability and ease of implementation. However, their main drawback is low efficiency, typically in the range of 30% to 45%, since the series pass element continuously dissipates power. Additionally, these supplies tend to be larger and heavier due to the use of line-frequency transformers and the need for significant heat dissipation [2], [3].

Furthermore, linear regulators can only step down the input voltage, requiring a higher unregulated input to produce a regulated output. They also lack inherent galvanic isolation unless an external transformer is used, and their performance is strongly influenced by the characteristics of the preceding rectification and filtering stages. These limitations make them less suitable for high-efficiency or high-power-density applications.

Switching Regulation: In switching power supplies, the regulating element operates as a high-frequency switch rather than a linear device. The input AC voltage is first rectified to an unregulated DC level, which is then rapidly switched to generate a high-frequency pulse train. This signal is processed through magnetic components and output filters to produce a regulated DC output, as illustrated in Figure 4:

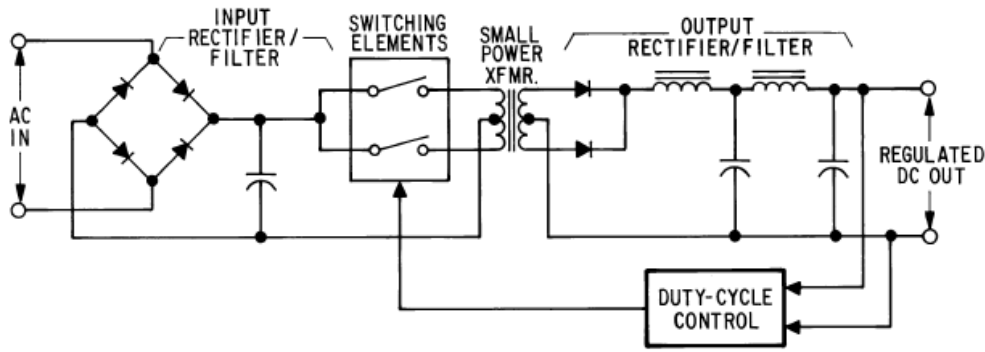


Fig. 4: Basic switching supply circuit [2]

Regulation is achieved by adjusting the duty cycle of the switching signal, allowing precise control of the energy delivered to the load. Since the switching devices operate predominantly in fully ON or OFF states, power dissipation within the control element is significantly reduced compared to linear regulators.

As a result, switching power supplies typically achieve higher efficiencies (commonly in the range of 65% to 85%) and enable substantial reductions in size and weight due to high-frequency operation. However, these advantages come at the cost of increased circuit complexity, switching noise, and electromagnetic interference (EMI), requiring careful design of filtering, control, and layout [2], [3].

High-Frequency Operation for Size Reduction: One of the main advantages of switched-mode power supplies is the ability to operate at high switching frequencies, which enables significant reductions in the size and weight of passive components. In particular, the required inductance and capacitance values decrease as switching frequency increases, allowing the use of smaller magnetic components and output filters.

This reduction directly improves power density, making high-frequency operation a key enabler for compact and lightweight designs. In contrast, linear power supplies rely on line-frequency transformers and large filtering components, resulting in bulkier implementations. As a result, modern power supply designs favor high-frequency switching to meet the increasing demand for miniaturization and efficiency [6], [3].

Switching Losses and Efficiency Limitations: Despite the advantages of high-frequency operation, increasing the switching frequency introduces additional losses that limit overall efficiency. These losses are primarily associated with the finite switching times of semiconductor devices, during which both voltage and current are present simultaneously, leading to switching dissipation.

Other significant contributions include gate driving losses, diode reverse recovery losses, and magnetic core losses, all of which tend to increase with frequency. As a consequence, there exists an optimal switching frequency that balances reduced passive component size with acceptable efficiency levels. In practical designs, this trade-off must be carefully evaluated to ensure that efficiency requirements are met without compromising power density [6], [5].

Parasitic Effects and Non-Idealities: At high switching frequencies, parasitic elements inherent to real components and circuit layouts become increasingly significant. These include parasitic inductances in PCB traces, parasitic capacitances in semiconductor devices, and non-ideal characteristics of passive components. Such parasitics can lead to voltage overshoots, ringing, increased electromagnetic interference (EMI), and additional power losses. For example, stray inductance in switching loops can cause high di/dt transitions, resulting in voltage spikes that stress semiconductor devices and degrade reliability.

Furthermore, non-ideal switching behavior, such as finite rise and fall times, contributes to increased switching losses and noise generation. These effects make careful PCB layout, component placement, and filtering essential aspects of high-frequency power supply design [7], [8].

III-C DC-DC Conversion Topologies

A switching regulator is a circuit that uses a controlled power switch, along with reactive elements such as inductors and capacitors, to transfer energy from input to output. The fundamental configurations include the buck (step-down), boost (step-up), and flyback converters, each offering specific advantages such as [9]:

- Output voltage greater or lower than the input.

- Galvanic isolation (in certain topologies as flyback).
- Reduced size and improved thermal performance due to high-frequency operation.

There are many additional DC-DC converter topologies; however, most are derived from these basic structures with modifications to improve performance, efficiency, or specific application requirements. Common examples include :

- Buck-boost converter.
- SEPIC (Single-Ended Primary Inductor Converter).
- Ćuk converter.
- Forward converter.
- Push-pull converter.
- Half-bridge and full-bridge converters.
- Resonant and quasi-resonant converters.

III-C1 Buck, Boost, and Flyback Converters

Figure 5 shows simplified electric diagrams of each topology. As it was established before, each circuit uses the same basic components located on different positions:

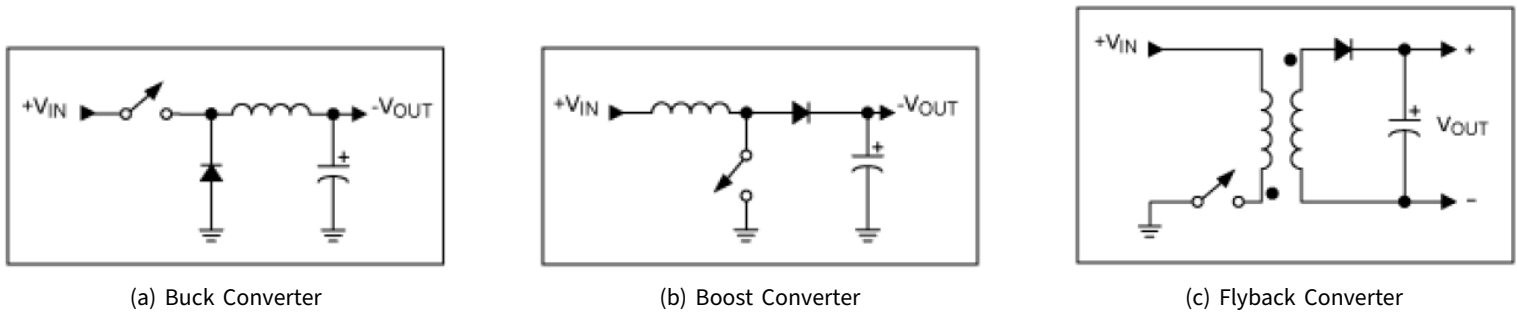


Fig. 5: Common DC-DC converter topologies: buck, boost, and flyback. [2]

In modern practical applications, DC-DC conversion circuits based on these topologies are significantly more complex. Mechanical or idealized switches are replaced by semiconductor devices such as transistors, typically driven by dedicated control ICs. These devices are regulated through additional feedback networks, which enable precise control of the output voltage, current, and overall system stability. Feedback and control circuitry are carefully nested around these circuits to regulate the energy transfer and maintain a constant output within normal operating conditions [9].

III-C2 Basics on Duty Control

From the derivations for the boost, buck, and inverter (flyback), it can be seen that changing the duty cycle controls the steady-state output with respect to the input voltage. This is a key concept governing all inductor-based switching circuits.

The most common control method, shown in Figure 6, is pulse-width modulation (PWM). In this approach, the output voltage is sampled and compared to a reference voltage to generate an error signal (V_{error}). This error signal is then compared with a ramp signal from an oscillator, and the comparator produces a PWM signal that drives the power switch. When the output voltage changes, V_{error} also changes, modifying the comparator threshold and, consequently, the duty cycle of the PWM signal. This adjustment drives the output voltage toward the desired value, reducing the error signal to zero and closing the control loop.

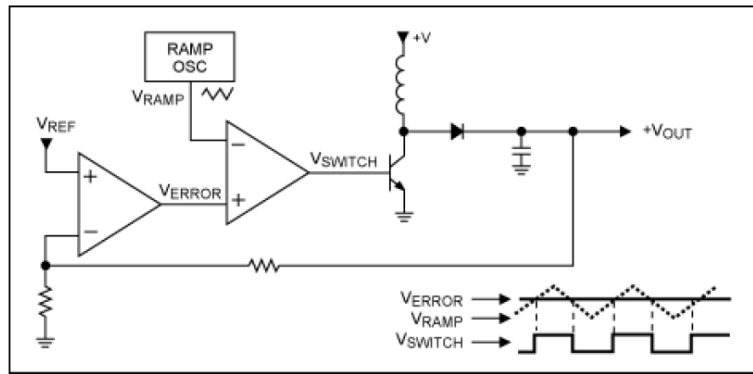


Fig. 6: Basic control loop [2]

III-C3 Use of Specialized Drivers and Integrated Circuits

Nowadays, a wide range of integrated circuits (ICs) are available to simplify the implementation of feedback control loops in switching power supplies. These devices integrate essential functions such as error amplifiers, voltage references, oscillators, PWM comparators, and gate drivers, reducing design complexity and improving reliability.

Depending on the application, some ICs also include protection features such as overcurrent, overvoltage, and thermal shutdown, as well as soft-start and compensation networks. This high level of integration enables more compact, efficient, and robust designs, while significantly reducing development time. Figure 7 disposes an block diagram of a IC circuit as example:

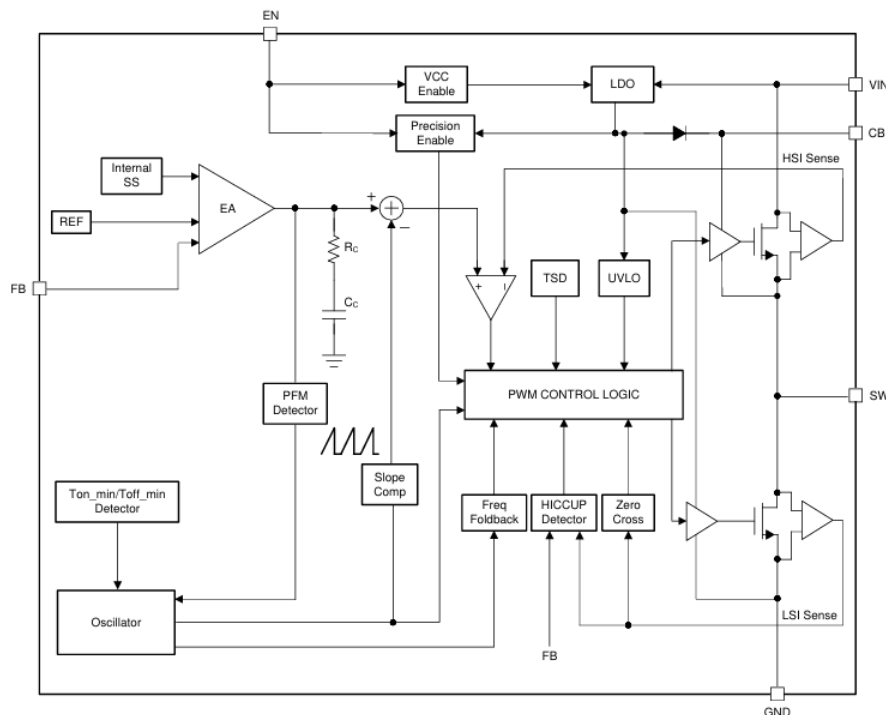


Fig. 7: LMR51430 block diagram circuit [10]

UCx84x Current-Mode PWM Controllers: The UCx84x family corresponds to current-mode PWM controllers typically used in isolated and non-isolated topologies such as flyback, forward, and buck converters. These controllers operate over a wide input voltage range and are commonly used in medium-power applications (tens to hundreds of watts). Current-mode control provides improved response and inherent cycle-by-cycle current limiting. Figure 8 shows a complete electric diagram of a Flyback Converter driven by an UC2842:

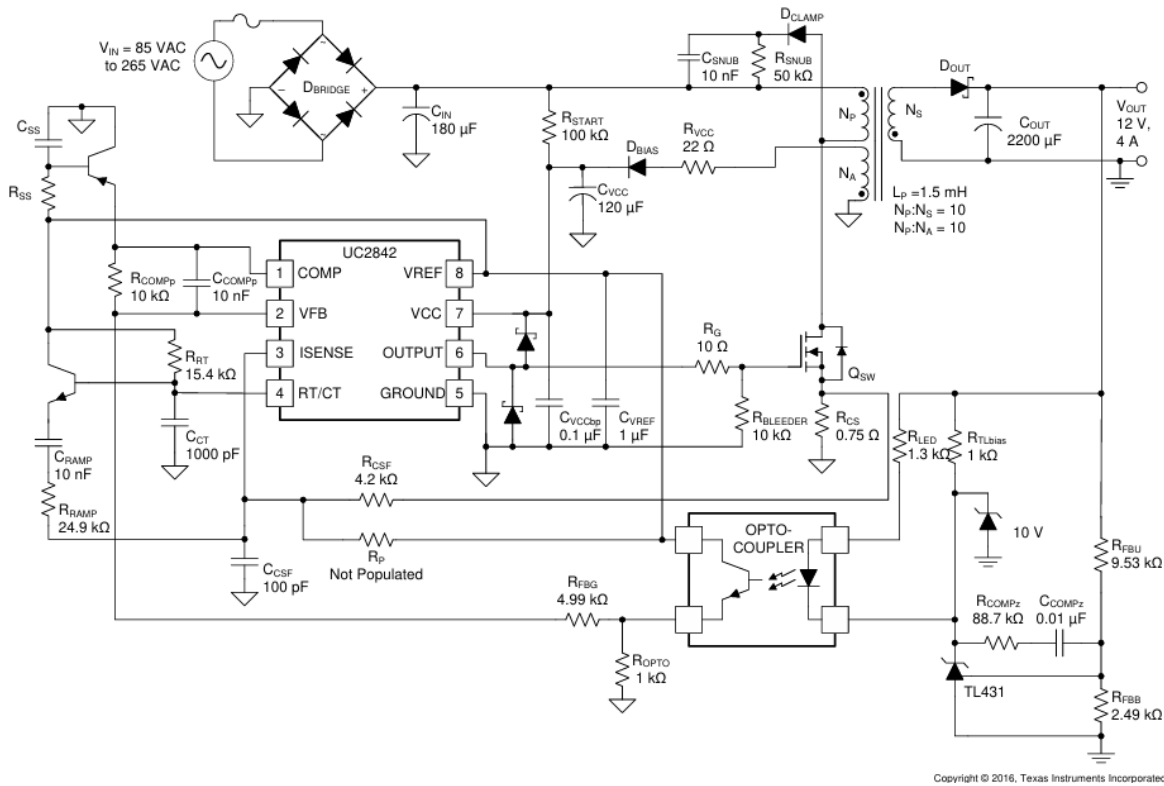


Fig. 8: Flyback topology circuit [11]

MAX1932 Boost Converter: The MAX1932 is a boost (step-up) converter with voltage-mode control, designed for applications requiring output voltages higher than the input. It integrates the switching control circuitry and operates over a broad range of input voltages, typically in low-to-medium power applications such as display biasing or portable systems.

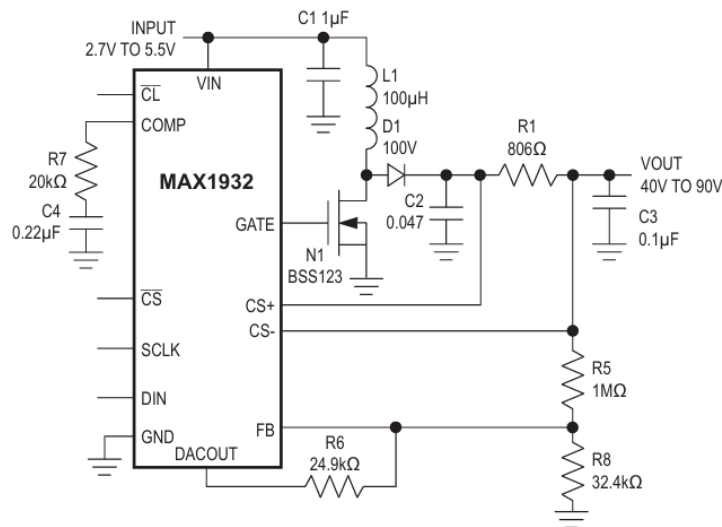


Fig. 9: MAX1932 boost converter diagram [12]

Although these integrated solutions appear significantly more complex, a detailed examination reveals that the fundamental converter topologies remain consistent with the basic configurations previously described. The additional circuitry mainly enhances control, protection, and stability, rather than altering the core energy transfer mechanism.

III-C4 Common Topologies in Laboratory Power Supplies

In modern laboratory power supplies, several converter topologies are commonly used depending on the required performance, power level, and cost constraints. For low-to-medium power ranges, the flyback converter is one of the most widely adopted topologies due to its simplicity, low component count, and inherent galvanic isolation. These characteristics make it particularly suitable for compact and cost-effective designs.

For higher power levels or improved efficiency and output performance, more advanced topologies such as forward, half-bridge, or full-bridge converters may be employed. These configurations offer better transformer utilization, lower stress on components, and improved efficiency, at the expense of increased circuit complexity.

In many laboratory-grade supplies, especially those requiring low noise and high precision, a hybrid approach is often used. A switching converter (commonly flyback or forward) provides initial voltage conversion and isolation, followed by a linear regulation stage to achieve fine control, low ripple, and fast transient response.

III-C5 Conduction Modes: CCM and DCM

Switching converters can operate under different conduction modes, which significantly influence their steady-state behavior, dynamic response, efficiency, and control complexity.

The two primary conduction modes are **Continuous Conduction Mode (CCM)** and **Discontinuous Conduction Mode (DCM)**. In CCM, the inductor or transformer current never falls to zero during a switching cycle. This results in lower current ripple, reduced peak currents, and improved efficiency at medium-to-high load conditions. However, CCM operation introduces more complex small-signal dynamics, including right-half-plane (RHP) zeros in certain topologies such as the flyback converter, which complicates loop compensation and stability design [3], [6].

In contrast, in DCM the inductor current falls to zero before the beginning of the next switching cycle. This mode simplifies the control loop dynamics, as it removes the RHP zero in flyback converters and results in a single-pole dominant response, making compensation significantly easier [4], [3]. However, DCM operation leads to higher peak currents, increased conduction losses, and higher output voltage ripple, which can reduce efficiency and increase stress on switching devices. Figure 10 shows graphs and equations for a buck converter in DCM and CCM:

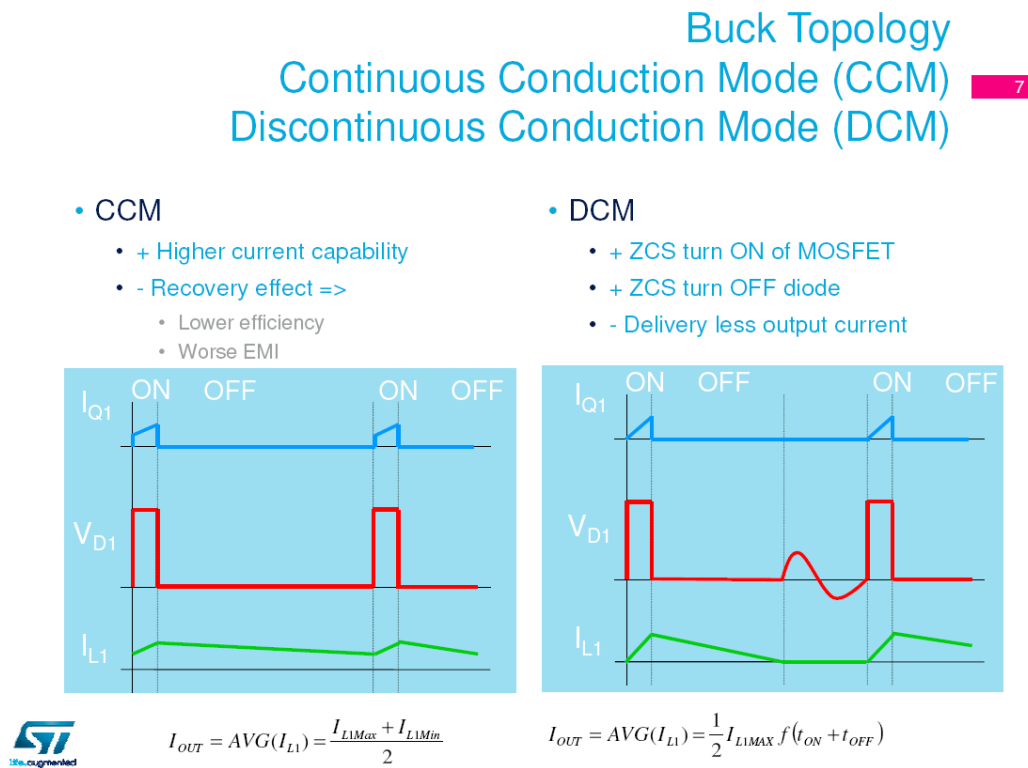


Fig. 10: Buck topology graphs and equations for CCM and DCM [6]

The selection between CCM and DCM represents a fundamental trade-off in power supply design. CCM is typically preferred in higher power applications due to its superior efficiency and lower RMS currents, while DCM is often used in low-power or light-load conditions due to its simpler control characteristics and reduced magnetic component size.

In practical implementations, many converters operate in a mixed or transitional regime depending on load conditions, commonly referred to as **Boundary Conduction Mode (BCM)** or critical conduction mode, where the inductor current reaches zero exactly at the end of each switching cycle. This mode offers a compromise between efficiency and control simplicity and is widely used in modern power factor correction (PFC) and flyback-based designs [6], [4].

III-D Control Strategies in Power Supplies

III-D1 Analog Control Techniques

Lineal Converter with Basic Control Loop: Analog control is fundamentally based on the principle of feedback. A variable of interest at the output is sensed and appropriately conditioned before being fed into a control stage. This processed signal is then used to drive an electronic switch or a dedicated driver, regulating the system in order to maintain the desired operating condition. Figure 11 presents a lineal AC-DC converter with a control loop:

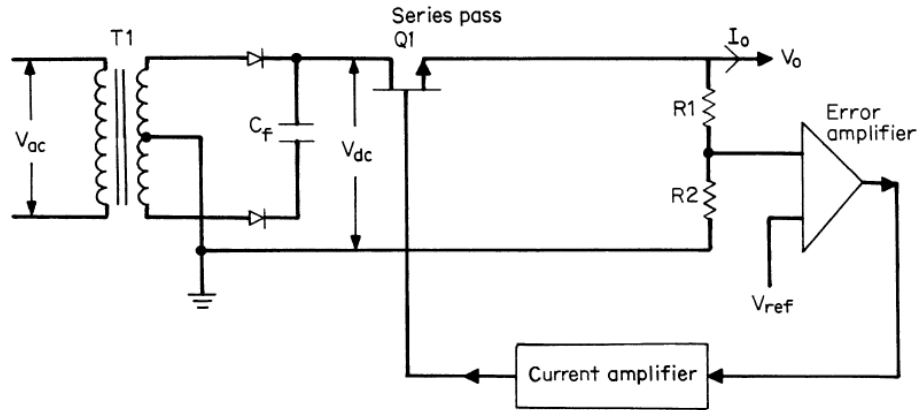


Fig. 11: Lineal AC-DC converter with feedback loop [3]

Any control loop is typically composed of a sensing network, an error amplifier, comparators, a voltage/current reference and a control or actuation stage. In general, any change in input voltage—due to, for example, AC input line voltage change, ripple, steady-state changes in the input or output, and any dynamic changes resulting from rapid load changes over its designed tolerance band—is absorbed across the series-pass element. This maintains the output voltage constant to an extent determined by the gain in the open-loop feedback amplifier [3].

Switching Converter with Basic Control Loop: Most switching converters primarily regulate the duty cycle of the power switch to control the transfer of energy from input to output. By adjusting the duty cycle, the converter directly influences the average output voltage or current.

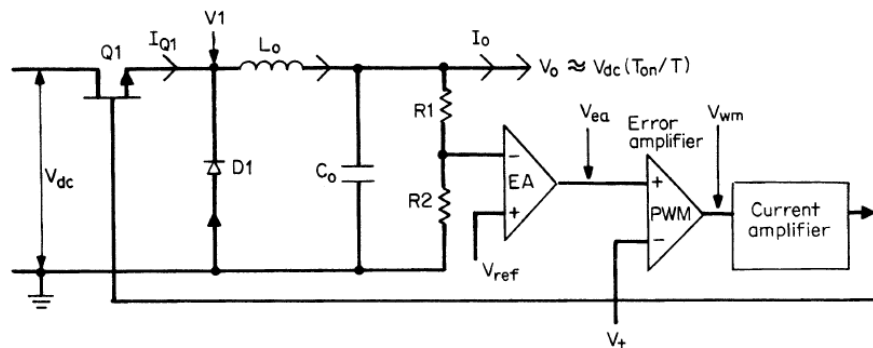


Fig. 12: Buck converter with feedback loop [3]

As Figure 12 shows, the control loop comes from a PWM Error Amplifier. As this modulation works exactly as the control logic proposed before. In practice, this basic control is complemented by additional control mechanisms depending on the topology and performance requirements. These may include current-mode control, voltage-mode compensation networks, slope compensation, and filtering techniques, all aimed at improving stability, reducing noise and ripple, and enhancing transient response.

More complex circuits integrate enhanced control loops with additional elements tailored to specific requirements. For example, Figure 13 shows a flyback DC-DC converter with a control loop that incorporates an optocoupler:

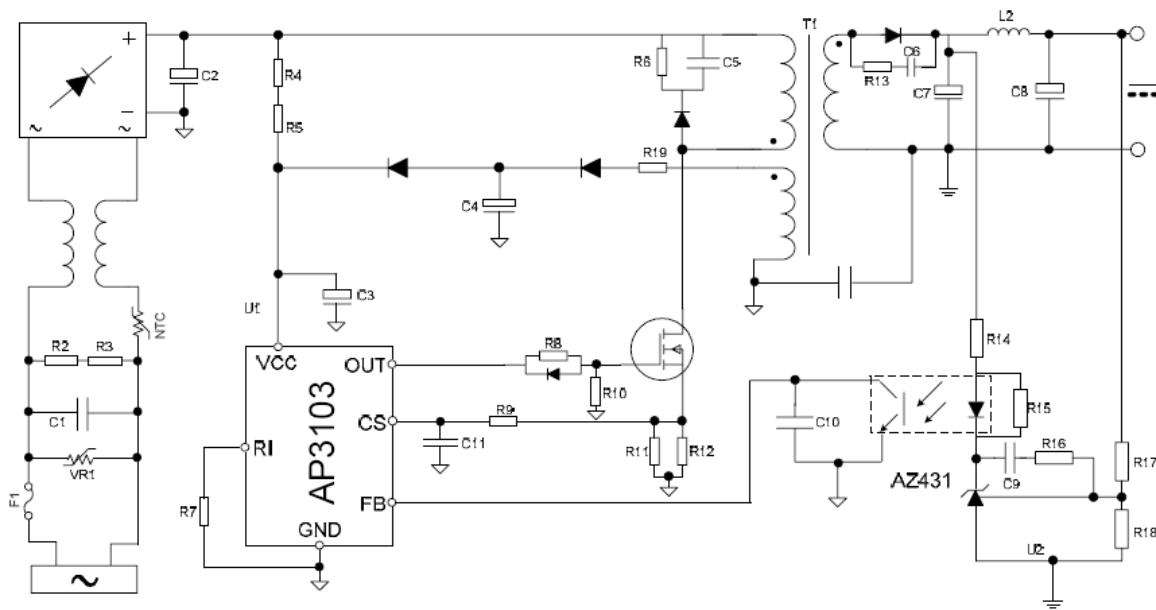


Fig. 13: System Schematic Circuit with AP3103 [13]

This addition allows galvanic isolation between the primary and secondary sides, enables accurate feedback from the output without compromising safety, improves regulation in isolated topologies, and reduces noise coupling between the input and output stages. Additionally, most circuits incorporate extra elements to enhance control and feedback; however, these additions are typically determined by the selection of a specific IC.

Types of Analog Control: In terms of control strategies, the most widely used are **voltage-mode control** and **current-mode control**. Voltage-mode control regulates the output by comparing the output voltage with a reference and adjusting the duty cycle accordingly. Current-mode control, on the other hand, introduces an inner current loop, providing faster response, improved stability, and inherent overcurrent protection, making it highly popular in modern designs.

Other variations and enhancements include hysteretic control, peak and average current-mode control, and constant on-time or off-time control, each selected based on specific requirements such as transient response, noise performance, and implementation complexity [6]. Figure 14 shows current mode principle of any circuit:

Current Mode Control – Principle

8

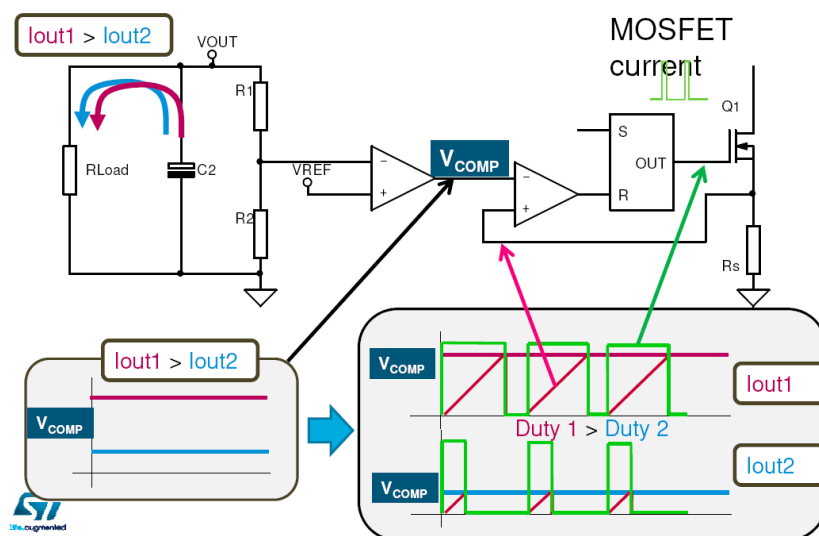


Fig. 14: Current mode principle [6]

Additionally, it should be noted that the selected conduction mode influences the choice of control strategy, which is determined according to the requirements for dynamic response, stability, and precision.

Conduction Mode and Control Strategy: The interaction between conduction mode and control strategy is fundamental in switching converter design. In CCM operation, converters exhibit second-order dynamics and, in certain topologies such as flyback, a right-half-plane (RHP) zero, which complicates loop compensation and stability design [6]. Current-mode control significantly simplifies this behavior by introducing an inner current loop, effectively reducing the system to a single-pole dominant response. This greatly facilitates loop compensation and improves transient response.

In contrast, DCM operation inherently presents a first-order system behavior, eliminating the dominant effects of the RHP zero. As a result, control design in DCM is naturally simpler, and the advantages of current-mode control are less critical compared to CCM operation [4], [6]. Therefore, current-mode control is particularly advantageous in CCM converters, where it mitigates the complexity of the system dynamics, whereas in DCM its impact is comparatively reduced [3], [4], [6].

III-D2 Digital Control Approaches

Two main approaches can be identified for the integration of digital control in power supplies: its use as the primary control loop, or as an auxiliary supervisory layer.

Digital loop as main control: In fully digital control architectures, the feedback loop is implemented using a digital processor, typically a microcontroller (MCU) or digital signal processor (DSP). In this case, analog signals such as output voltage and current are acquired through analog-to-digital converters (ADCs), processed by a control algorithm (PID or state-based control), and used to generate a control signal through pulse-width modulation (PWM) or digital-to-analog converters (DACs). This approach enables high flexibility, programmability, and advanced control features such as adaptive tuning, non-linear control strategies, and system monitoring.

Digital loop as secondary control: Alternatively, digital control can be implemented as a supervisory or auxiliary layer, while the primary regulation loop remains analog. In this hybrid or mixed-signal approach, the fast inner control loop (typically voltage-mode or current-mode) is handled by analog circuitry, while the digital system is responsible for setpoint generation, user interface, protection management, and communication. This structure preserves the high bandwidth and low latency of analog control while adding flexibility and configurability through digital processing [14], [15].

Type of Digital Control selected: Digital control offers several advantages, including ease of reconfiguration, integration with user interfaces, and implementation of complex control algorithms. However, it also introduces limitations such as sampling delays, quantization effects, and reduced control bandwidth compared to purely analog implementations. These constraints must be carefully considered, particularly in high-frequency switching converters where fast dynamic response is required.

As a result, hybrid control architectures are widely adopted in practical designs, combining the robustness and speed of analog control with the versatility of digital systems. In this case, digital control is not implemented as the primary regulation loop but rather as a supervisory layer. The main control loop responsible for voltage and current regulation is implemented using analog circuitry, ensuring high bandwidth, low latency, and stable operation.

The digital subsystem, based on a microcontroller, is used for setpoint generation, user interaction, monitoring, and system management. This approach corresponds to a mixed-signal control architecture, where the advantages of both analog and digital domains are combined. This structure avoids the complexity associated with fully digital control, such as sampling delays, quantization effects, and reduced control bandwidth, while still providing flexibility and programmability through the digital interface [15].

III-E Constant Voltage and Constant Current Operation

III-E1 Definitions

Compliance Voltage: Compliance voltage is the maximum output voltage a power supply can provide while maintaining a set constant current. When this limit is reached, the supply can no longer regulate the current.

Constant Current Power Supply: A constant current (CC) power supply maintains a fixed output current by adjusting its output voltage within its compliance limit. If the required voltage exceeds this limit, current regulation is lost.

Constant Voltage Power Supply: A constant voltage (CV) power supply maintains a fixed output voltage while allowing the current to vary with the load. If the current exceeds the maximum rating, the supply leaves voltage regulation and enters current-limited operation.

Constant Voltage / Constant Current (CV/CC) Power Supply: A CV/CC power supply operates as a **constant voltage source for relatively high load resistance** and as a **constant current source for relatively low load resistance**. The automatic transition between these modes occurs at a critical (crossover) load resistance defined as Equation 3 indicates:

$$R_c = \frac{V_s}{I_s} = \frac{E_s}{I_s} \quad (3)$$

where V_s is the set output voltage and I_s is the set current limit. Figure 15 shows a graph with the output characteristic of this modes:

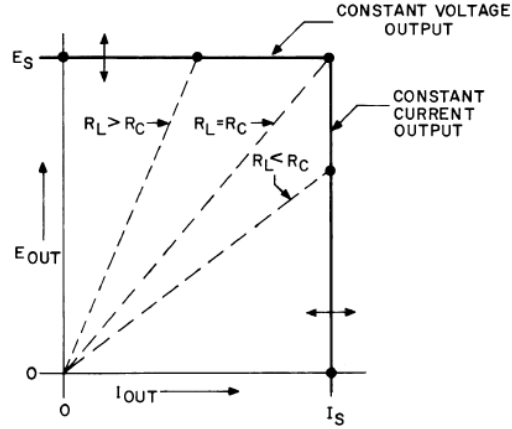


Fig. 15: CV/CC Output characteristic [2]

III-E2 Practical Implementation of CV/CC

CV/CC circuit on Lineal Regulated Converter: First, a simplified analysis of the operation of these modes is proposed in a linear-operation converter. Figure 16 presents a circuit diagram of an AC-DC converter including voltage and current sensing and regulation loops:

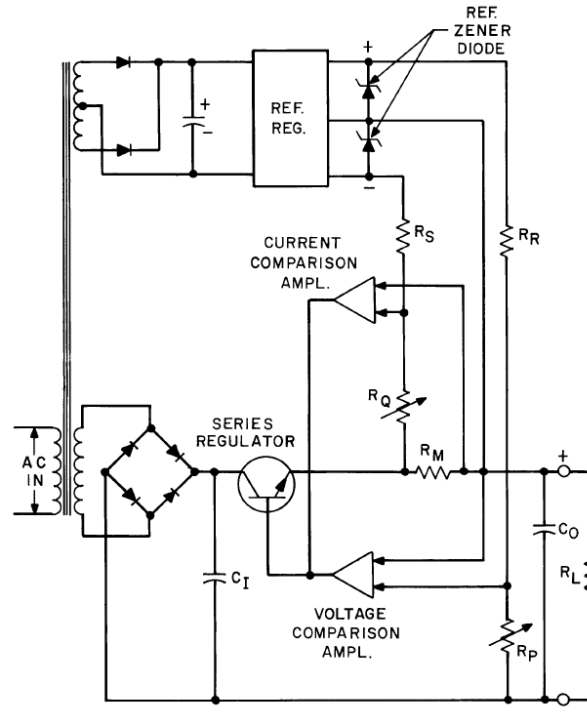


Fig. 16: Constant Voltage/Constant Current CV/CC Power Supply [2]

In this circuit, the selection of the output voltage and current limit is achieved through a combination of a fixed reference stage and adjustable feedback networks.

First, a stable reference voltage is generated by the reference regulator and Zener diodes. This reference serves as the baseline for both voltage and current regulation loops.

For voltage regulation (CV mode), the output voltage is sensed through a resistive divider composed of R_R and the adjustable resistor R_P . This network feeds a fraction of the output voltage back to the comparison stage. By varying R_P , the division ratio changes, modifying the sensed voltage. The system regulates the output such that the sensed voltage equals the reference voltage, thereby setting the desired output level.

For current regulation (CC mode), the output current is sensed through the voltage developed across the series sensing network, where R_M acts as the main current-sensing resistor. The voltage drop associated with the output current can be expressed as $V_{RM} = I_{out} \cdot R_M$. This sensed voltage is then processed through a network involving R_S and the adjustable resistor R_Q , which together establish the effective reference threshold for current limiting. In this configuration, R_S and R_Q form a scaling and adjustment network that defines the level at which the current comparator is triggered. When the conditioned sensing voltage reaches this threshold, the control loop limits the current, enforcing constant-current operation.

Both regulation mechanisms act simultaneously, but only one dominates at a given time. Under normal conditions, the voltage loop controls the output (CV mode). When the load current reaches the predefined limit set by the R_S - R_Q network, the current loop takes control (CC mode), reducing the output voltage as needed to maintain the set current. It should be noted that the comparison blocks shown in the diagram are not simple ideal comparators, but rather threshold-based comparison stages typically implemented using operational amplifiers. Their internal circuitry is omitted in the diagram for clarity, as their detailed implementation is not essential for understanding the overall control principle.

Transition Between Modes: The transition between these two modes is achieved automatically by appropriate decoupling within the control circuitry. This transfer occurs at a critical value of load resistance, defined by the ratio between the programmed output voltage and the programmed current limit $R_C = \frac{V_{set}}{I_{set}}$. Figure 17 shows the operating locus of the supply:

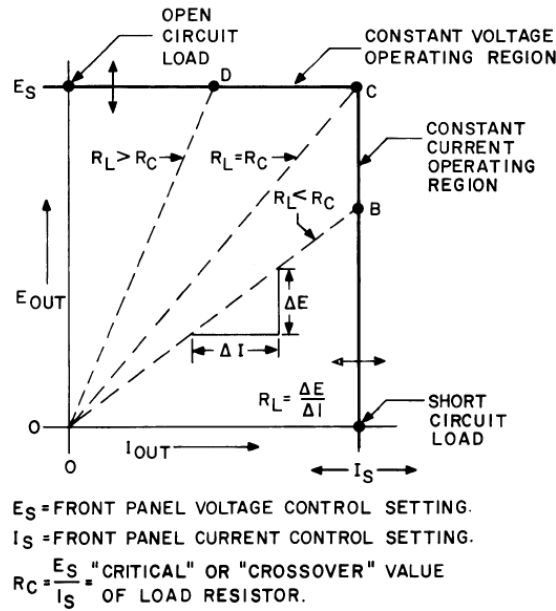


Fig. 17: Operating locus of a CV/CC supply [2]

For either constant voltage or constant current operation, the proper choice of E_S and I_S insures optimum protection for the load device as well as full protection for the power supply. The slope of the line connecting the origin to any operating point on the locus shown in Figure 17 is proportional to the value of the load resistance connected to the output terminals of the supply. The critical, or crossover, load resistance is defined as $R_C = \frac{E_S}{I_S}$, where E_S and I_S are the voltage and current control settings, respectively.

By adjusting these controls, the crossover resistance can be set to any desired value within the operating range. If the load resistance R_L is greater than R_C , the supply operates in constant voltage (CV) mode. Conversely, if R_L is less than R_C , the supply operates in constant current (CC) mode.

Digital Implementation: According to [16], digital implementation of a CV/CC power supply introduces significant advantages in terms of flexibility, configurability, and system-level integration compared to purely analog approaches. In particular, digital control enables the decoupling of voltage and current reference generation from the primary control loop, allowing real-time programmability of operating points and dynamic adaptation to different load conditions.

III-F Protection Strategies and Input Safeguards

III-F1 Integrated Protection: TPS2660 eFuse

The TPS2660 is an integrated eFuse device designed to provide robust protection at the input stage of a power supply [17]. It incorporates multiple protection mechanisms, including overcurrent protection (OCP), overvoltage protection (OVP), undervoltage lockout (UVLO), inrush current control, and thermal shutdown within a single device [18]. Figure 19 depicts a block working diagram of this IC.

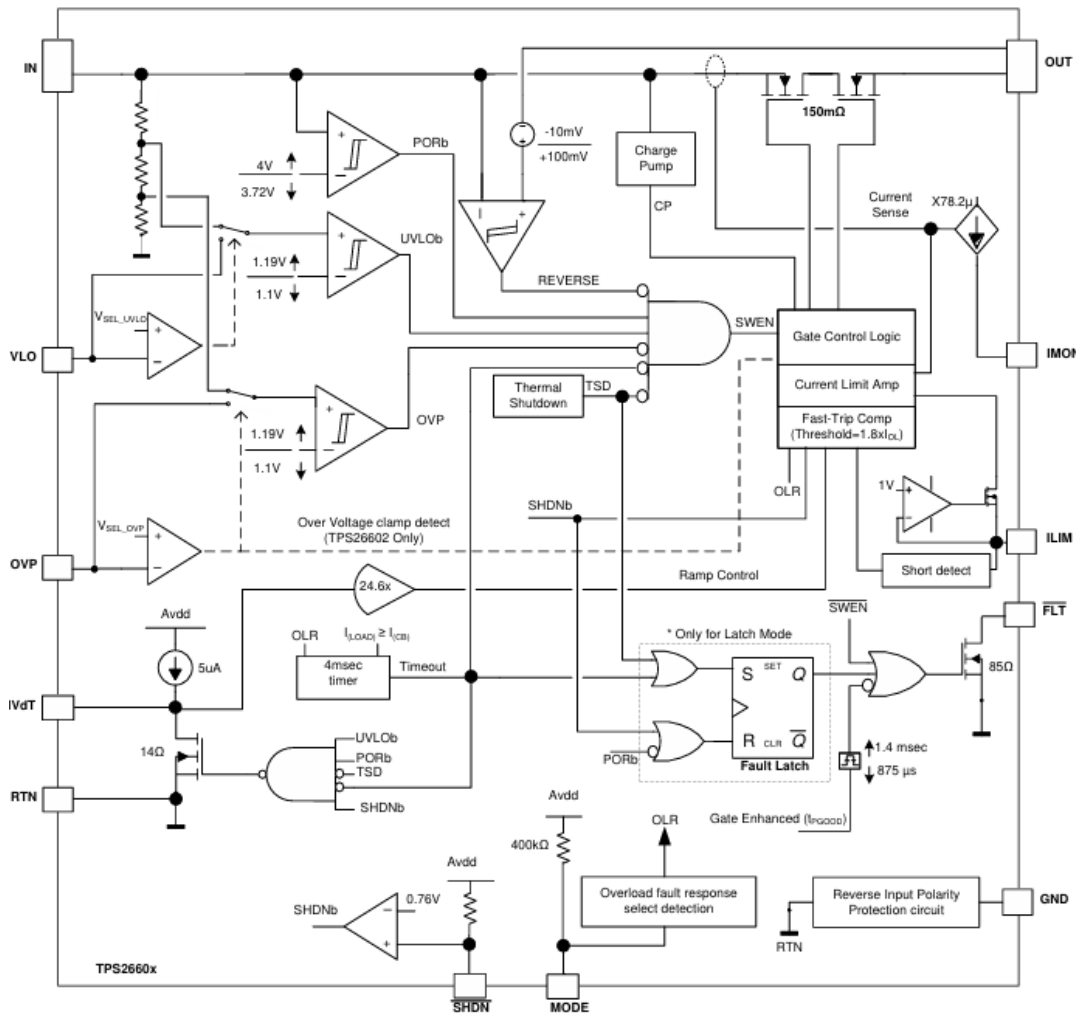


Fig. 19: Functional block diagram of TPS2660 [17]

Functionally, it operates as a smart high-side switch that disconnects the load under fault conditions. Its programmable current limiting and fast transient response significantly enhance system reliability, preventing damage to downstream power conversion stages [18]. This level of integration simplifies the design process and ensures predictable protection behavior under a wide range of operating conditions.

III-F2 Discrete Protection Implementation

A discrete protection approach is typically implemented using a combination of components such as fuses, MOSFETs, sense resistors, comparators or operational amplifiers, and transient voltage suppression (TVS) devices [19], [20]. These elements are arranged to provide basic protection functions like overcurrent limiting, overvoltage clamping, and fault isolation.

While this method offers flexibility and lower component cost, it requires careful design to ensure proper coordination between protection mechanisms, stability under transient conditions, and reliable fault response [3].

III-F3 Comparison of Approaches

Compared to discrete implementations, integrated solutions such as the TPS2660 provide a higher level of reliability, reduced design complexity, and faster protection response due to internal optimization and tight parameter control [18]. In contrast, discrete designs

offer lower cost and greater customization but at the expense of increased development effort, larger PCB area, and potentially less predictable behavior under fault conditions [3]. Consequently, the choice between both approaches depends on the target application, with integrated solutions being more suitable for laboratory-grade and industrial systems, while discrete implementations may be sufficient for cost-sensitive or less critical designs.

III-G Power Factor Correction

III-G1 Power Factor Fundamentals and Impact on Rectification

In AC-DC power supplies, the rectification stage introduces non-linear current draw from the mains, resulting in a low power factor and significant harmonic distortion. The power factor (PF) is defined as the ratio between real power and apparent power, and is degraded when the input current is not sinusoidal or not in phase with the input voltage. In conventional diode bridge rectifiers followed by bulk capacitors, current is drawn in short pulses, leading to poor power factor and increased stress on the electrical network [4], [21].

III-G2 Relevance in Laboratory Power Supplies

Although power factor correction is traditionally associated with high-power or grid-connected systems, it has become increasingly relevant in modern laboratory power supplies due to regulatory requirements and efficiency considerations. Standards related to harmonic emissions and standby consumption have driven the adoption of PFC techniques even in medium-power applications. However, for moderate power levels, such as the one considered in this project, the implementation of PFC is not always mandatory, and its inclusion depends on design constraints, cost, and compliance requirements [4], [22].

III-G3 Power Factor Correction Techniques

Power factor correction techniques aim to shape the input current waveform to follow the input voltage, thereby improving the power factor and reducing harmonic distortion. These techniques can be broadly classified into passive and active methods. Passive methods rely on line-frequency inductors and capacitors, while active methods use high-frequency switching converters, typically based on boost topologies, to actively control the input current. Active PFC is widely adopted in modern switched-mode power supplies due to its superior performance and reduced size [4], [6].

Given the scope of this project, a detailed analysis of PFC implementation is beyond the objectives of this proposal. However, the following aspects should be reviewed in the literature if PFC integration is considered:

- Impact of rectification on input current waveform and harmonic distortion.
- Regulatory standards related to power factor and harmonic emissions.
- Passive versus active PFC topologies and their trade-offs.
- Control strategies for active PFC (e.g., current shaping and average current-mode control).
- Efficiency and thermal implications of adding a PFC stage.

III-G4 Passive vs Active PFC

Passive PFC techniques are based on bulky inductors and capacitors operating at line frequency, which limits their effectiveness and increases size and weight. In contrast, active PFC employs high-frequency switching converters to achieve near-unity power factor and significantly improved harmonic performance. While active PFC provides better efficiency and compliance with modern standards, it introduces additional complexity in control, switching losses, and electromagnetic interference considerations. As a result, the selection between passive and active PFC depends on power level, regulatory requirements, and system constraints [4], [7].

III-H Electromagnetic Interferences

This section is dense in both theoretical background and design considerations; therefore, only the relevant references are provided for review in the next iteration of this document:

- [8] – *EMI/EMC in Switching Power Supplies* (Analog Devices)
- [23] – *EMI Filter Design for Power Supplies* (Texas Instruments)
- [24] – *EMI Filter Design* (Ozenbaugh)
- [25] – *Electromagnetic Compatibility Engineering* (Ott)
- [26] – *EMC for Product Designers* (Williams)
- [27] – *EMC Filter Design Guide* (Würth Elektronik)

III-I Modern Trends in Power Supply Design

III-I1 Integration and Modular Design Approaches

Modern power supply design increasingly relies on high levels of integration, where control, protection, and power management functions are incorporated into dedicated integrated circuits. These devices often include pulse-width modulation (PWM) controllers, current sensing, and compensation networks, significantly reducing design complexity and improving reliability. In addition, modular design approaches based on reference designs and power modules enable faster development cycles and predictable performance, which are particularly valuable in laboratory and industrial applications [14], [21].

III-I2 Efficiency Optimization and Standby Performance

Efficiency optimization has become a critical requirement in modern power supplies, not only at full load but also under light-load and standby conditions. Regulatory standards have driven the adoption of techniques such as burst-mode operation, pulse-skipping, and adaptive control strategies to minimize power consumption when the load demand is low. These approaches allow power supplies to maintain high efficiency across a wide operating range while complying with energy efficiency requirements [28], [22].

III-I3 Trends in Switching Frequency

A key trend in power electronics is the gradual increase of switching frequency to reduce the size of passive components and improve power density. At the same time, variable switching frequency techniques are increasingly employed to optimize efficiency under varying load conditions and to mitigate electromagnetic interference (EMI). These strategies reflect the trade-off between efficiency, thermal performance, and electromagnetic compatibility in modern power supply design [6], [21].

IV Requirements and Constraints

To ensure the proper planning and development of the project, it is essential to establish clear definitions for the prototype's requirements and constraints. For the purposes of this section, a requirement refers to what the prototype is expected to achieve or perform, and may be negotiable depending on project priorities. Conversely, a constraint refers to a mandatory limitation or restriction that must be respected during the design and development process, and is therefore considered non-negotiable.

TABLE II: Project requirements and constraints matrix

ID	Category	Type	Requirement / Constraint Statement	Justification	VT
E1	Electrical	Req.	The equipment shall provide an adjustable DC output voltage from 0 to 30 V.	This range was selected from a commercial comparison of general-purpose laboratory DC power supplies.	B
E2	Electrical	Req.	The equipment shall provide an adjustable output or limited current from 0 to 3 A.	This range was selected from a commercial comparison of low-power laboratory DC power supplies.	B, H
E3	Electrical	Req.	The equipment shall provide a minimum continuous output power of 90 W.	This value is obtained from the maximum specified output condition: 30 V and 3 A.	F, I
E4	Electrical	Req.	The voltage and current measurement resolution shall be at least 10 mV and 5 mA, respectively.	These values were defined from the expected ADC resolution and the required measurement granularity for laboratory use.	B
E5	Electrical	Req.	The output ripple shall not exceed 50 mV peak-to-peak.	This limit was selected as a practical quality criterion for a low-cost laboratory DC source.	C
E6	Electrical	Req.	When the load changes, the output voltage variation shall remain below 2%.	This value was selected from commercial load-regulation specifications for bench DC power supplies.	D
E7	Electrical	Req.	The equipment shall be able to operate in constant-voltage and constant-current modes.	These modes are standard operating functions in laboratory DC power supplies.	B
E8	Electrical	Req.	The power conversion efficiency shall be at least 50%.	This value was defined as a minimum feasible target for the selected low-cost power topology.	F
E9	Electrical	Req.	The equipment shall operate without changes beyond 2% for voltage and 1% for the current for multiple hours.	The voltage must not change through the working period in a functional source.	J
E10	Electrical	Req.	The equipment shall operate with a power factor close to 1.	This criterion was included to reduce input-current distortion and improve AC-side power quality.	F
E11	Electrical	Const.	The input voltage shall be 120 V RMS at 60 Hz, with a tolerance margin of $\pm 10\%$.	This condition corresponds to the nominal AC supply available in the intended laboratory environment.	E
E12	Electrical	Const.	The equipment shall not present functional failures caused by electromagnetic interference in a laboratory environment.	This constraint is aligned with the EMC immunity approach of IEC 61000-6 for electronic equipment environments.	–
M1	Mechanical	Req.	The equipment shall not weigh more than 6 kg.	This limit was selected from a commercial comparison of portable bench DC power supplies.	L
M2	Mechanical	Req.	The equipment shall have external dimensions smaller than 20 cm $\times$ 20 cm $\times$ 30 cm.	These dimensions were selected from a commercial comparison of compact bench DC power supplies.	L
M3	Mechanical	Const.	The equipment shall include gripping areas intended to make transportation easier.	This constraint supports single-user handling of a portable laboratory instrument.	L
M4	Mechanical	Const.	The enclosure shall support the total weight of the equipment and preserve its structural integrity during normal handling and transportation, without permanent deformation, accidental opening, or exposure of internal components.	This condition supports the mechanical protection expected for enclosed electrical laboratory equipment.	A, L
T1	Thermal	Const.	The internal temperature of the equipment shall not exceed 70 °C after 20 minutes of operation.	This limit was selected to keep internal components below typical maximum operating-temperature margins.	I
T2	Thermal	Const.	The equipment shall operate properly within an ambient temperature range from 0 °C to 25 °C.	This range was defined from the expected indoor laboratory operating conditions.	I
S1	Safety	Const.	User interaction with energized circuit elements shall be reduced to a minimum.	The internal components of the source should be inaccessible to the user for safety concerns.	A
S2	Safety	Const.	The user shall not be able to access the inside of the equipment without the use of tools.	The internal components of the source should be inaccessible to the user for safety concerns.	A
S3	Safety	Const.	The equipment shall include a physical protective earth connection, corresponding to Class I insulation.	The earth connection reduces the risks of electrical shock.	A
S4	Safety	Const.	The equipment shall include protection against overvoltage and overcurrent conditions.	The overvoltage and overcurrent real hazards that have to be taken into account.	A, G, H
S5	Safety	Const.	The input and output stages shall be electrically isolated.	The insulation of the output is a general good practice for electronic devices.	A
S6	Safety	Const.	The power supply shall withstand a short circuit at the output without permanent damage.	This requirement supports safe operation under foreseeable laboratory misuse conditions.	H
I1	User Interaction	Req.	The output shall be turned on and off from the user interface.	This function is based on standard operation of commercial programmable bench power supplies.	G, K
I2	User Interaction	Req.	The user shall be able to modify the output voltage and current values through an interface.	This is a basic control function required for adjustable laboratory DC power supplies.	B, K
I3	User Interaction	Req.	The interface shall display the output voltage and current variables in real time.	This requirement supports direct monitoring during laboratory tests.	B, K
I4	User Interaction	Req.	The fuse shall be accessible for replacement without requiring complete disassembly of the power supply.	This criterion improves maintainability after an input protection event.	H
I5	User Interaction	Req.	The user shall be able to quickly install the equipment on a workbench.	This requirement is based on typical plug-and-use operation of bench laboratory instruments.	K
I6	User Interaction	Req.	The measurement displayed on the screen shall correspond to the actual output values with an error of ± 5 mV.	This value was defined from the expected voltage measurement accuracy of the monitoring stage.	B
I7	User Interaction	Req.	The user interface shall be intuitive enough for an engineering student to understand its use after reviewing the manual.	This criterion matches the intended educational use of the equipment.	K
I8	User Interaction	Const.	The required input connections shall be standard and easily available.	This constraint reduces cost and improves compatibility with common laboratory accessories.	K

V Data Management

As the device does not manage relevant information and does not save any data, data management is irrelevant for this practical implementation.

VI Controlled Testing

The controlled tests of the prototype are design to test the correct implementation of the thermal, electrical, mechanical, safety and user experience requirements and constraints. The tests are proposed as follow:

VI-A Pre-operation inspection

Objective: Verify if the prototype can be energized in a safe way before testing the other functions.

Tested requirements and constraints: S1, S2, S3, S4, S5, M4

Required equipment: multimeter, screwdriver.

Procedure:

- 1) Inspect the enclosure, output terminals, input connector, fuse holder, grounding connection, internal wiring, PCB fixation, and cable routing.
- 2) Verify that there are no loose conductors, exposed AC terminals, sharp edges, unsecured components, or accessible energized parts.
- 3) Check that the enclosure cannot be opened without tools.
- 4) Verify continuity between the protective earth terminal and the metallic enclosure.

Success criteria: The equipment does not present visible mechanical or electrical risks. Internal energized parts are not be accessible during normal operation. The enclosure remains closed unless tools are used. The protective earth connection is present and continuous.

VI-B Output Range and Setpoint Verification

Objective: Verify that the power supply can provide the required output voltage and current ranges.

Tested requirements and constraints: E1, E2, E4, E7, I2, I3, I6

Required equipment: multimeter x2, resistive load (rheostat)

Procedure:

- 1) Connect the load to the terminals of the source.
- 2) Configure the output voltage of the source to 0 V, 5 V, 10 V, 12 V, 18 V, 24 V and 30 V, for each of the setpoints measure the output voltage and the current with the multimeter.
- 3) Change the source to C.C and configure the output current of the source to 0.1 A, 0.5 A, 1 A, 1.5 A, 2 A, 2.5 A and 3 A. For each of the setpoints measure the output current and the voltage with the multimeter.
- 4) Compare the measured values with the ideal values and the ones displayed on the screen of the device.

Success criteria: The equipment covers the full 0–30 V and 0-3 A range. The output measured remains less than 10 mV or 5 mA away from the displayed output and ideal value. The interface updates as the values are changed.

VI-C Ripple Verification

Objective: Verify the quality of the DC output by measuring output ripple.

Tested requirements and constraints: E5

Required equipment: oscilloscope, probes, load,

Procedure:

- 1) Connect the load to the terminals of the source.
- 2) Configure the output voltage of the source to 2.5 V, 5 V, 10 V, 12 V, 18 V, 24 V and 30 V, for each of the setpoints measure the ripple voltage through the oscilloscope, the voltage is measure peak to peak.

Success criteria: For the whole range the ripple stays below 50 mV.

VI-D Load Regulation Verification

Objective: Verify that the output voltage remains stable when the load changes.

Tested requirements and constraints: E6

Required equipment: multimeter, load (rheostat)

Procedure:

- 1) Connect the load to the terminals of the source and set the voltage to 12 V.
- 2) Set the load resistance to obtain an output current of approximately 0.3 A.
- 3) Set the load resistance to obtain an output current of approximately 2.7 A.
- 4) Calculate the load regulation as $\frac{|V_{high} - V_{low}|}{V_{set}} * 100$.
- 5) Repeat previous steps for 24 V.

Success criteria: The load regulation stays under 2%.

VI-E Input Voltage Range and Line Regulation Verification

Objective: Verify that the power supply operates correctly when the AC input voltage varies within the specified range.

Tested requirements and constraints: E10

Required equipment: multimeter, load, AC source

Procedure:

- 1) Connect the load to the terminals of the source and set the output to 24 V and 2 A.
- 2) Set the input at 108 V RMS and measure the output voltage.
- 3) Set the input at 120 V RMS and measure the output voltage.
- 4) Set the input at 132 V RMS and measure the output voltage.
- 5) Calculate the line regulation as $\frac{|V_{high} - V_{low}|}{V_{set}} * 100$.

Success criteria: The line regulation stays under 2%.

VI-F Output Power, Efficiency, and Power Factor Verification

Objective: Verify output power capability, conversion efficiency, and input power factor.

Tested requirements and constraints: E3, E8, E9

Required equipment: multimeter x2, load, power analyzer (supposed)

Procedure:

- 1) Connect the load to the terminals of the source and set the output to 5 V and 1 A.
- 2) Measure the output voltage and current with the multimeter.
- 3) Calculate the output power as $P_{out} = V_{out} I_{out}$
- 4) Calculate the efficiency as $\frac{P_{out}}{P_{in}} \times 100\%$
- 5) Calculate the power factor as.....
- 6) Repeat the tests for 12 V, 24 V and 30 V.

Success criteria: The source delivers 90 W under nominal operating conditions. The efficiency is at least 50% in all cases.

VI-G Start-Up and Shutdown Behavior Verification

Objective: Verify that the power supply behaves safely during start-up, shutdown, and output enable/disable actions.

Tested requirements and constraints: S4, I1

Required equipment: oscilloscope, load, multimeter

Procedure:

- 1) Configure the output at 24 V with the load connected.
- 2) With the oscilloscope measure the output voltage during AC power-on. Record maximum overshoot, undershoot, start-up time and shut-down behavior.
- 3) With the oscilloscope measure the output voltage during AC power-off. Record maximum overshoot, undershoot, start-up time and shut-down behavior.

- 4) With the oscilloscope measure the output voltage during output enable. Record maximum overshoot, undershoot, start-up time and shut-down behavior.
- 5) With the oscilloscope measure the output voltage during output disable. Record maximum overshoot, undershoot, start-up time and shut-down behavior.

Success criteria: The output voltage shall not exceed 110% of the configured voltage during start-up, shutdown, output enable, or output disable.

VI-H Protection and Fault Response Verification

Objective: Verify that the protection functions operate correctly under abnormal but foreseeable laboratory conditions.

Tested requirements and constraints: E2, S4, S6, I4

Required equipment: load, multimeter

Procedure:

- 1) Verify the current-limiting behavior by increasing the load current until the configured limit is reached, confirm that it does not surpass it.
- 2) Verify the behavior under a controlled output short-circuit condition, using appropriate laboratory supervision and current-limited test conditions. After the fail conditions, return the equipment to normal operation and confirm that it's working correctly.
- 3) Verify the fuse replacement procedure.

Success criteria: The current does not exceed the configured current limit during current-limiting operation. The equipment remains functional after the controlled short-circuit test. The fuse is accessible for replacement without complete disassembly.

VI-I Thermal Performance and Endurance Verification

Objective: Verify that the equipment can operate under representative high-power conditions without exceeding the thermal limit.

Tested requirements and constraints: T1, T2, E3

Required equipment: infrared thermometer, load, stopwatch, multimeter

Procedure:

- 1) Operate the equipment on nominal output for 30 minutes.
- 2) Every 5 minutes measure the enclosure surface temperature using an infrared thermometer and measure internal temperature using a thermocouple if available.

Success criteria: The internal temperature shall not exceed 70 °C. The equipment does not shut down unexpectedly, lose regulation, restart, or show signs of thermal damage.

VI-J Drift Verification

Objective: Verify that the equipment can operate for a long period of time without affecting the equipment itself.

Tested requirements and constraints:

Required equipment: load, multimeter

Procedure:

- 1) Operate the equipment at an output of 15 V and 1.5 V for 8 hours.
- 2) Take measurements of the output voltage and current every 30 minutes.

Success criteria: During the whole process the output of voltage does not change more than $\pm 2\%$ and the current does not change more than $\pm 1\%$.

VI-K User Interaction and Usability Verification

Objective: Verify that the intended user can operate the power supply correctly after reviewing the manual.

Tested requirements and constraints: I1, I2, I3, I5, I7, I8

Required equipment: phone survey

Procedure:

- 1) Ask students in the electronics laboratories to help with the survey. At least 10 students should be surveyed.
- 2) The student will carry the equipment from one side to the hallway to the other one.
- 3) The student will be asked to connect the equipment in one of the benches.
- 4) The student will be asked to do some basic tests with the source after reading the manual.
- 5) The student will be asked to answer a survey grading how easy is to carry the equipment, how easy is to connect, how easy is to use.

Success criteria: The average score for each survey item shall be greater than or equal to 6/10, and at least 80% of users shall complete the basic operation tasks without direct assistance.

VI-L Mechanical and Enclosure Verification

Objective: Verify that the equipment satisfies its mechanical requirements.

Tested requirements and constraints: M1, M2, M3, M4

Required equipment: scale, measuring tape

Procedure:

- 1) Measure the total mass of the equipment.
- 2) Measure the dimensions of the equipment.
- 3) After finishing all the previous tests, inspect visually the state of the enclosure, looking for deformation, loose panels, exposed components, etc.

Success criteria: The equipment weighs less than 6 kg. The dimensions are smaller than 20 cm × 20 cm × 30 cm. The enclosure remains in good conditions after all the handling.

VII Deployment

This section provides a detailed description of how the system will be deployed once the prototype has been validated through controlled testing. After the validation stage, the project will be released as an open-source initiative, with the purpose of giving it a more accessible, transparent, and educational character. The deployment of the system is planned through a public GitHub repository. This repository will contain the necessary technical documentation, design files, source code, validation protocols, and general project information required for other students, researchers, or electronics enthusiasts to understand, reproduce, modify, and improve the prototype. In this way, the project is not intended only as a functional laboratory power supply, but also as an educational resource for learning about electronic design, power regulation, prototyping, and validation procedures.

Since the project will be deployed as an open-source development, there are no plans for large-scale distribution or commercial production. The objective is not to convert the prototype into a market-ready product, but to provide a documented and reproducible design that can be used for academic, experimental, and learning purposes. Therefore, the deployment strategy focuses on accessibility, documentation quality, and ease of replication rather than manufacturing, certification, logistics, or commercial support. The GitHub repository will serve as the main platform for publication and future maintenance of the project. It may include sections such as the system requirements, design decisions, electrical schematics, PCB files, mechanical design files, firmware, user manual, testing procedures, and validation results. Additionally, the repository may allow other users to report issues, suggest improvements, or propose modifications through version-control tools and collaborative development practices.

In summary, the system will be deployed as an open-source educational prototype through GitHub. This approach supports the free distribution of knowledge, encourages collaborative improvement, and allows the project to remain available for future academic or personal developments without the need for commercial deployment.

Ewigvolt Open-Source Repository

The complete project documentation, including design files, schematics, PCB resources, firmware, mechanical files, testing procedures, and future updates, will be made available through the official GitHub repository. This repository will serve as the main access point for students, researchers, and electronics enthusiasts interested in studying, reproducing, modifying, or improving the Ewigvolt power supply prototype.

github.com/juamorenogo/DCTSource_DIY

VIII Development Stages Diagram

The Development Stages Diagram summarizes the structured process followed for the development of the Ewivolt adjustable CV/CC DC power supply prototype. This diagram was defined using a deliverable-oriented approach, in which the project is divided into sequential phases that cover the complete development cycle: project definition, system architecture, electronic and control design, mechanical design, prototype integration, verification and validation, and open-source deployment. By organizing the project in this way, each stage can be associated with specific tasks, expected outputs, and verification activities, supporting a clearer and more traceable development process.

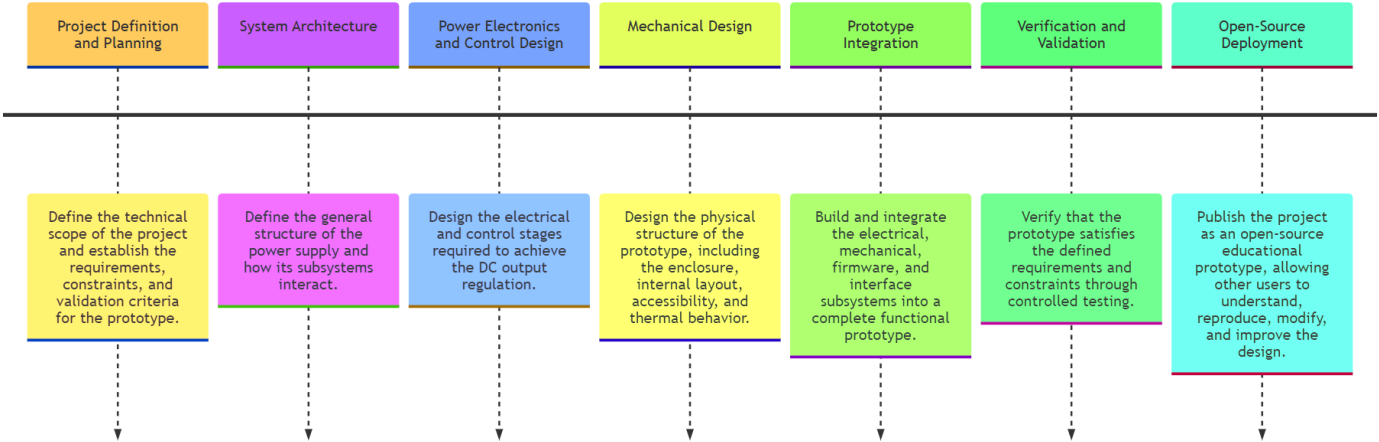


Fig. 20: Caption

VIII-A Project Definition and Planning

The purpose of this phase is to define the technical scope of the project and establish the initial planning criteria required for the development of the Ewivolt prototype. During this stage, the main problem addressed by the project is identified, together with the motivation for designing an open-source adjustable DC laboratory power supply. This phase also defines the functional expectations of the prototype, including its electrical, mechanical, thermal, safety, and user interaction requirements. The outcome of this phase is a clear requirements and constraints framework that guides the rest of the design process. The requirements describe what the prototype is expected to achieve, while the constraints define the non-negotiable limitations that must be respected during development. In addition, this phase establishes a traceability relationship between each requirement and its corresponding verification test, ensuring that the final prototype can be evaluated through objective validation criteria.

TABLE III: Phase 1 task-level breakdown.

Task	General description
Problem statement	Describes the technical and educational problem that motivates the development of an open-source adjustable DC laboratory power supply.
Project motivation	Explains the relevance of the project in terms of affordability, usability, design transparency, reproducibility, and educational value.
Requirements definition	Establishes the expected functions and performance targets of the prototype, including voltage range, current range, power, ripple, regulation, and user interaction.
Constraints definition	Defines mandatory design limitations related to safety, isolation, thermal behavior, enclosure design, input voltage, and user protection.
Requirement traceability	Links each requirement or constraint with the corresponding verification test, supporting a structured validation process.
Validation plan	Defines the general testing strategy required to verify whether the prototype satisfies the established requirements and constraints.

VIII-B System Architecture

This phase defines the general structure of the power supply and describes how the different subsystems interact to achieve the desired operation. The system architecture divides the prototype into its main functional blocks, including the AC input stage, input protection, power conversion stage, output regulation stage, sensing circuits, control unit, user interface, safety protections, and mechanical enclosure. The purpose of this architectural division is to simplify the design process and provide a clear view of the role of each subsystem. The functional architecture also supports the identification of interfaces between subsystems, such as the connection between voltage and current sensors and the control stage, or the interaction between the user interface and the output regulation system. At this stage, the operating modes of the device are also defined, especially constant-voltage and constant-current operation, as well as the general safety and usability concepts of the prototype.

TABLE IV: Phase 2 task-level breakdown.

Task	General description
Functional architecture	Defines the main functions that the power supply must perform as an adjustable CV/CC laboratory instrument.
Device block diagram	Represents the prototype as a set of interconnected functional blocks, including AC input, protection, power conversion, sensing, control, user interface, and output stage.
Subsystem definition	Divides the device into smaller subsystems that can be designed, analyzed, integrated, and tested separately.
Subsystem interfaces	Defines how the electrical, control, user interface, protection, and mechanical subsystems interact with each other.
Safety concept	Defines the general safety approach of the system, including isolation, grounding, enclosure protection, and fault response.
User interface concept	Defines how the user will configure voltage and current values, monitor the output variables, and control the activation of the source.

VIII-C Electronic Design

This phase covers the detailed electrical and control design required to obtain a regulated DC output. It includes the design of the AC input stage, input protection, isolation strategy, power converter, output filtering, voltage sensing, current sensing, control loops, and protection circuits. Since the prototype is intended to operate as an adjustable laboratory DC power supply, this phase is directly related to the main electrical performance targets of the project. The design must allow the equipment to provide an output voltage range from 0 to 30 V, an output current range from 0 to 3 A, and a minimum output power of 90 W. In addition, the control strategy must support both constant-voltage and constant-current modes, while maintaining low ripple, adequate load regulation, and stable operation under different load conditions. This phase also includes considerations related to electromagnetic interference, loop compensation, schematic design, and PCB layout, since these aspects directly affect the stability, safety, and reliability of the final prototype.

TABLE V: Phase 3 task-level breakdown.

Task	General description
AC input stage	Defines the input section connected to the AC supply and the conditions required for safe operation at the specified input voltage.
Input protection	Includes the selection or design of protection elements intended to reduce risks caused by abnormal input conditions.
Isolation stage	Provides electrical separation between the AC input side and the low-voltage output and control sections.
Power converter	Defines the main power conversion stage used to generate the adjustable DC output required by the prototype.
Output stage	Conditions the converted power before it is delivered to the output terminals, considering filtering, stability, and load connection.
Voltage sensing	Measures the output voltage so that it can be used by the control system and displayed to the user.
Current sensing	Measures the output current to support monitoring, current limiting, and constant-current operation.
CC/CV control	Implements the control strategy required to regulate the output in constant-voltage and constant-current modes.
Protection circuits	Includes overcurrent, overvoltage, short-circuit, and related protection functions required for safe operation.
EMI considerations	Considers layout, grounding, filtering, switching effects, and electromagnetic compatibility aspects that may affect the operation of the prototype.
Loop compensation	Adjusts the control response to improve stability and reduce the risk of oscillations during load or setpoint changes.
Schematic design	Produces the complete electrical diagram of the prototype, including power, control, sensing, interface, and protection circuits.
PCB design	Converts the schematic into a manufacturable printed circuit board layout, considering routing, component placement, thermal behavior, and electrical safety.

VIII-D Mechanical Design

The mechanical design phase defines the physical structure of the prototype, including the enclosure, internal component layout, mounting points, accessibility, ventilation, and thermal dissipation strategy. The main purpose of this phase is to ensure that the equipment is compact, portable, safe for the user, and suitable for laboratory use. During this stage, the enclosure must be designed to protect the internal electrical components and prevent direct access to energized parts during normal operation. The internal layout must allow proper mounting of the PCB, power components, connectors, fuse holder, display, control interface, and wiring. Special attention must also be given to heat dissipation, since the prototype must operate under representative load conditions without exceeding the defined thermal limit. Finally, the mechanical design must comply with the dimensional and weight restrictions established for the project, while also including handling features that make transportation and installation easier.

TABLE VI: Phase 4 task-level breakdown.

Task	General description
Enclosure design	Defines the external housing of the prototype, protecting internal components and preventing direct user access to energized parts.
Internal layout	Organizes the internal components to improve assembly, accessibility, wiring, ventilation, and thermal behavior.
Component mounting	Defines how PCBs, connectors, power components, display, controls, and protection elements will be fixed inside the enclosure.
Thermal dissipation	Considers heat sinks, component placement, and internal airflow to keep the equipment within the defined thermal limits.
Handling features	Includes grips, accessible surfaces, or structural features that make the device easier to carry and install on a workbench.

VIII-E Prototype Integration

The prototype integration phase transforms the electrical, mechanical, firmware, and interface designs into a complete functional prototype. This stage includes component procurement, PCB manufacturing, electronic assembly, firmware upload, mechanical assembly, subsystem prototyping, and final integration of all components. Before energizing the system, a pre-operation inspection must be carried out to verify that the prototype does not present visible mechanical or electrical risks. This inspection includes checking the grounding connection, internal wiring, PCB fixation, cable routing, enclosure closure, fuse accessibility, insulation, and absence of loose conductors or exposed terminals. The objective of this phase is not only to assemble the prototype, but also to ensure that the integrated system is ready for controlled testing under safe initial conditions.

TABLE VII: Phase 5 task-level breakdown.

Task	General description
Component procurement	Identifies and obtains the electronic, mechanical, and interface components required to build the prototype.
PCB manufacturing	Produces the printed circuit board based on the final PCB design files.
Electronic assembly	Mounts and solders the electronic components on the PCB or auxiliary boards.
Firmware upload	Loads the control and interface firmware into the selected microcontroller or digital control platform.
Mechanical assembly	Builds the enclosure and installs internal supports, connectors, interface elements, and mounting features.
Subsystem prototyping	Tests individual subsystems before combining them into the complete prototype.
Electronic integration	Connects the power stage, sensing circuits, controller, interface, and protection elements into a functional electrical system.
Mechanical integration	Installs the assembled electronic system inside the enclosure while verifying spacing, fixation, accessibility, and cable routing.
Final assembly	Combines all subsystems into a complete functional prototype ready for inspection and controlled testing.

VIII-F Verification and Validation

This phase verifies that the prototype satisfies the defined requirements and constraints through controlled testing. The validation process includes electrical, thermal, mechanical, safety, and user interaction tests. Each test is associated with specific requirements from the project matrix, allowing the performance of the prototype to be evaluated in a traceable and objective way. The verification activities include output range and setpoint validation, ripple measurement, load regulation testing, input voltage and line regulation verification, output power and efficiency evaluation, power factor verification, start-up and shutdown behavior analysis, protection and fault response testing, thermal endurance testing, mechanical inspection, and usability evaluation. The result of this phase is a validation report that determines whether the prototype meets the expected performance criteria and identifies any required design corrections or improvements before final release.

TABLE VIII: Phase 6 task-level breakdown.

Task	General description
Pre-operation inspection	Verifies that the prototype can be energized safely before performing functional tests.
Controlled Testing Implementation	Performs the tests of the controlled testing section with the appropriate equipment and cautions.
Validation report	Summarizes the test results, compliance status, limitations, and required design corrections or improvements.

VIII-G Open Source Deployment

The final phase consists of publishing the project as an open-source educational prototype. The objective is not only to develop a functional DC laboratory power supply, but also to provide a documented and reproducible design that can be studied, modified, and improved by other users. The deployment will be carried out through a public GitHub repository containing the hardware files, firmware, mechanical design files, testing procedures, validation results, user manual, README file, license information, and final documentation. This repository will serve as the main platform for distributing the project and supporting future improvements. By releasing the design as an open-source project, Ewigvolt becomes a practical educational tool for learning about power electronics, control, PCB design, prototyping, and validation of electronic systems.

TABLE IX: Phase 7 task-level breakdown.

Task	General description
GitHub repository organization	Structures the repository so that documentation, hardware files, firmware, mechanical files, and validation results are easy to locate.
Hardware files publication	Publishes schematics, PCB files, bills of materials, and other hardware-related files required to reproduce the design.
Firmware publication	Publishes the source code used for control, measurement, protection, and user interface functions.
Mechanical files publication	Publishes enclosure models, drawings, or manufacturing files required to reproduce the mechanical design.
Testing procedures publication	Provides the validation procedures so that other users can evaluate their own implementation of the prototype.
Validation results publication	Shares the obtained test results to document the real performance of the prototype.
User manual release	Provides basic instructions for installation, configuration, operation, maintenance, and safe use.
README file	Introduces the project, explains its purpose, lists the repository contents, and guides new users through the documentation.
License definition	Defines the legal conditions under which the project may be used, modified, and redistributed.
Final documentation release	Publishes the complete project documentation as the final open-source deliverable.

IX Device Block Diagram

IX-A Summary of the Proposed Design and System Block Diagram

The main design decisions adopted for the proposed system are summarized in Table X.

TABLE X: Summary of Design Decisions

Aspect	Selected Approach
Converter Topology	DC-DC Flyback
Primary Control	Analog CV/CC dual-loop control
Supervisory Control	Digital (microcontroller-based)
Isolation Method	Transformer + optocoupler feedback
PWM / Driving	Dedicated flyback controller / driver IC
Current Sensing	Dedicated current-sense amplifier IC
Voltage Sensing	Resistive divider with analog conditioning
Input Protection	Discrete protection circuitry
Output Protection	Dedicated output protection stage
Auxiliary Supplies	Separate analog and digital rails
Reference Generation	Precision voltage reference
EMI Mitigation	Input EMI filter
Power Factor Correction	Optional PFC stage

The design choices summarized in Table X are reflected in the system-level architecture shown in Figure 21. The diagram illustrates the interaction between the power stage, the analog control loops, and the digital supervisory layer. In particular, it highlights the separation between the primary (high-voltage) and secondary (isolated) domains, as well as the feedback path implemented through the optocoupler. The dual-loop CV/CC control structure is explicitly represented, showing how voltage and current sensing feed independent error amplifiers whose outputs are combined through a dominant-loop selection mechanism to drive the PWM modulator. Additionally, auxiliary subsystems such as sensing, protection, and reference generation are integrated into the architecture, providing a comprehensive view of the proposed implementation.

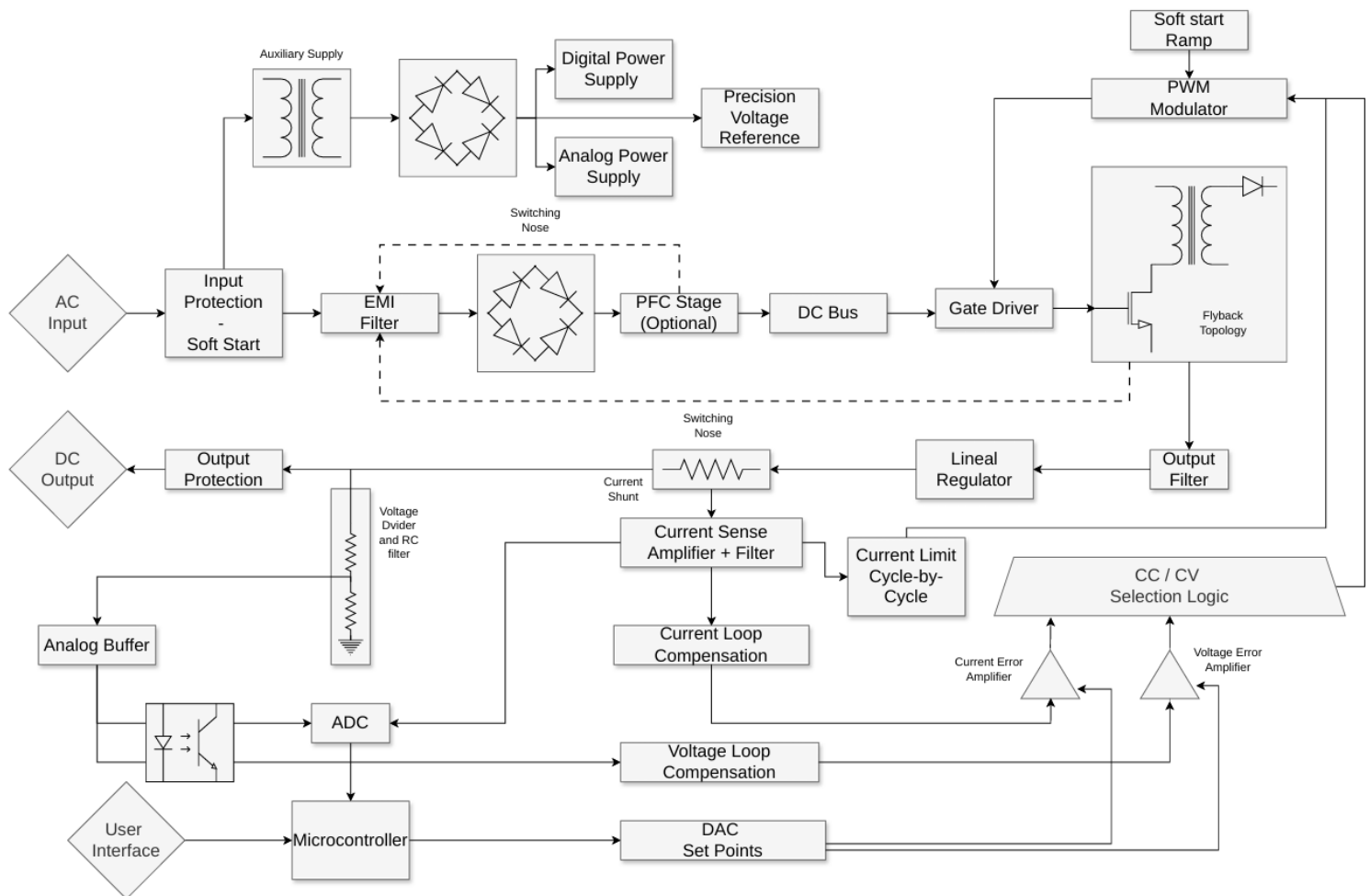


Fig. 21: Block diagram of the proposed system

The proposed system is based on a flyback DC-DC converter topology, selected due to its simplicity, reduced component count, and inherent capability to provide galvanic isolation between the input and output stages. This isolation is achieved through the transformer and reinforced in the feedback path via the use of an optocoupler, ensuring safe operation when interfacing high-voltage input stages with low-voltage outputs.

The regulation strategy is implemented using a dual-loop analog control scheme, where constant voltage (CV) and constant current (CC) operation are achieved through independent error amplifiers. A dominant-loop selection mechanism determines which loop governs the system at any given time, ensuring smooth transition between CV and CC modes. This analog control structure constitutes the primary regulation loop, providing high bandwidth and fast dynamic response.

In parallel, a digital control layer based on a microcontroller is introduced as a supervisory system. This layer is responsible for generating voltage and current setpoints through a DAC, acquiring measurements via an ADC, and handling user interaction. The separation between the fast analog control domain and the slower digital domain improves system robustness, minimizes noise sensitivity, and avoids stability issues associated with fully digital control loops.

The power stage is driven using a dedicated flyback controller or gate driver IC, which handles PWM generation and switching control. Input protection is implemented using discrete circuitry to provide flexibility in handling fault conditions such as inrush current and overvoltage. On the sensing side, current measurement is performed using a dedicated current-sense amplifier, while voltage sensing is implemented using a resistive divider with analog filtering, balancing accuracy and simplicity.

Additional features such as soft-start and cycle-by-cycle current limiting are incorporated at the hardware level to enhance system reliability and protect the switching elements under transient and fault conditions. Finally, EMI considerations are addressed through the inclusion of an input filter and identification of switching noise sources, while an optional PFC stage is considered for future compliance with regulatory requirements.

IX-B Digital Interface Subsystem

The figure 23 shows the basic digital blocks required for the construction of the interface:

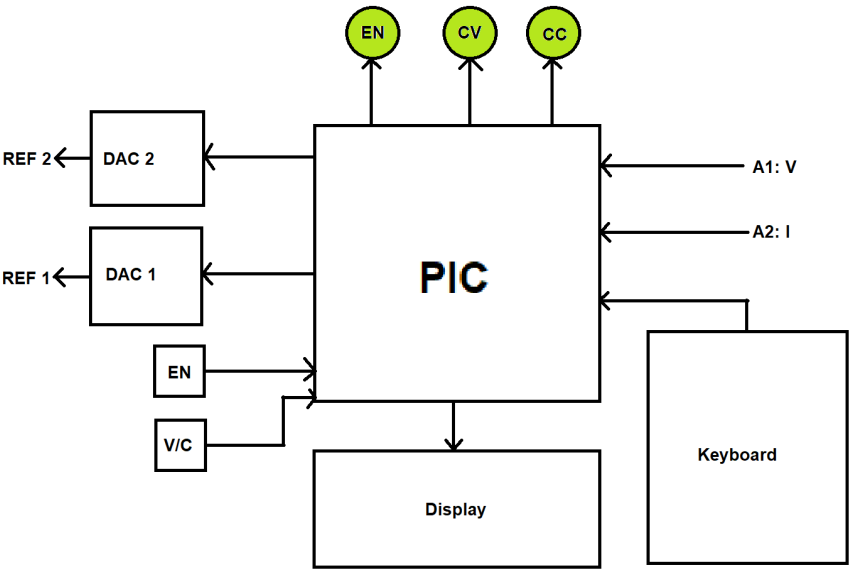


Fig. 22: Digital interface block diagram

The main block is the micro-controller, which processes all input and output signals associated with the interface. It sets the operating points through two DACs, using the values entered by the matrix keypad. The micro-controller also receives the voltage and current measurements through the ADCs and displays these values on a 20x4 LCD screen. The interface also includes two push buttons: EN, used to enable the output, and V/C, used to switch between CV and CC operation modes. The system status is indicated by three LEDs: one for the enabled output, one for CV mode, and one for CC mode.

X Practical Considerations

At this stage, the following topics are not developed as full subsections. Instead, they are presented as a structured list of aspects that must be reviewed and validated in future iterations of the design. Many of these considerations depend on practical implementation, particularly PCB layout and hardware testing, and are therefore intentionally deferred while still being identified due to their relevance.

- **General Considerations**

- Thermal management and power handling
- Efficiency and load conditions
- Component selection criteria

- **EMI and EMC Considerations**

- Sources of electromagnetic interference
- Conducted and radiated emissions
- Filtering and noise mitigation techniques

- **PCB Design and Layout Guidelines**

- High di/dt current loops and routing
- Grounding techniques and return paths
- Decoupling and component placement strategies

XI Schematic and PCB Design

At this stage, the schematic and PCB design are not presented as finalized implementations. Instead, this section outlines the main design blocks and considerations that will guide the development of the hardware. The proposed schematics will be based on previously validated reference designs and application notes, such as [29], [30], incorporating modifications to meet the specific requirements of the system.

In several cases, different implementation approaches will be evaluated, including the use of dedicated integrated circuits versus discrete component solutions. The final selection will be based on trade-offs between complexity, performance, cost, and flexibility.

- **Protection and Safety Mechanisms**

- Overvoltage and overcurrent protection (crowbar circuits, current foldback)
- Input protection and surge suppression
- User safety and isolation techniques

- **Power Stage Schematic**

- AC input stage and rectification
- Input protection implementation
- DC-DC conversion stage (flyback topology)

- **Control and Regulation Implementation**

- Voltage regulation circuitry
- Current sensing and limiting implementation
- CV/CC control interaction and loop integration

- **Loop Compensation and Stability**

- Compensation network design
- Stability considerations and design trade-offs

- **EMI Considerations in Hardware**

- Noise sources in the power stage
- Input and output filtering implementation
- Switching noise mitigation techniques

- **PCB Design and Layout**

- Power and signal separation strategies
- High di/dt loop minimization
- Grounding strategy and return paths
- Decoupling and component placement
- Thermal layout considerations

Additionally, practical design guidelines and layout techniques specific to flyback converters, such as those discussed in [6], will be considered during the PCB implementation phase.

XI-A Digital Interface Implementation

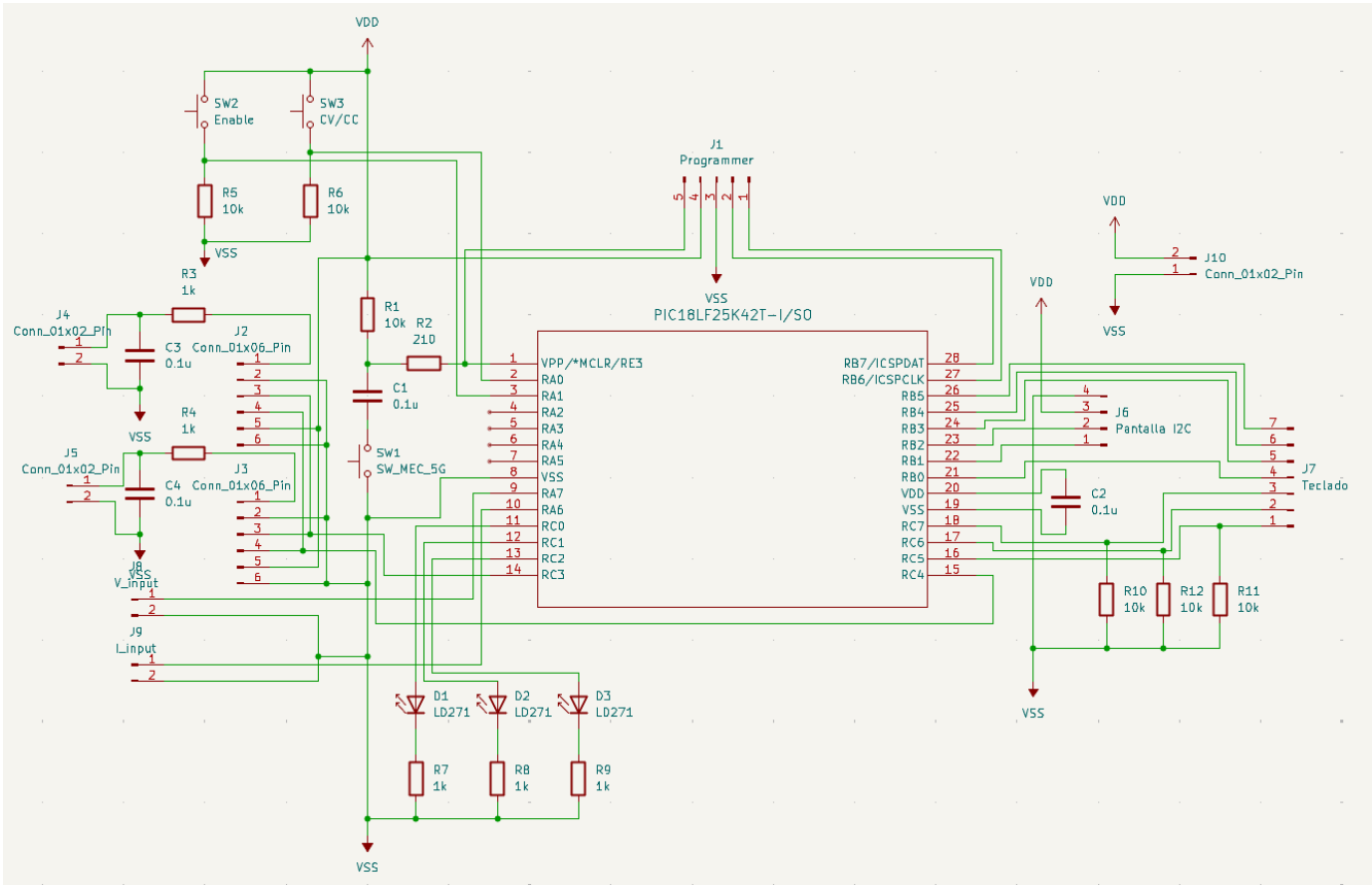


Fig. 23: Digital interface block diagram

XI-A1 Microcontroller Integration

The digital interface schematic implements the user interaction and supervision stage of the Ewigvolt DC power supply. This stage is based on a PIC18LF25K42 microcontroller, which manages the user inputs, the display, the digital-to-analog converters, and the basic control signals required to define the operating point of the source. The interface is designed to support voltage and current setpoint configuration, real-time visualization, output enable control, and interaction with the analog regulation stage. The microcontroller is powered from a regulated 3.3 V digital supply. This voltage level is selected because the PIC18LF25K42 belongs to the low-voltage family of the device. Therefore, the I/O signals, I2C communication lines, keypad inputs, and digital control outputs are referenced to the same 3.3 V logic level. Decoupling capacitors are placed close to the microcontroller power pins in order to reduce local supply noise and improve the stability of the digital circuit.

XI-A2 DAC-Based Setpoint Generation

The digital interface uses two external DACs because the internal resources of the microcontroller do not provide the resolution and analog output quality required to generate precise voltage and current reference signals. For this reason, the selected device is the MCP4725, a 12-bit digital-to-analog converter that communicates through the I2C protocol. Each DAC is used to generate one independent analog reference: one for the voltage setpoint and the other for the current limit setpoint. The specific MCP4725 module used in this design is a compact breakout board that allows its I2C address to be modified through minor hardware adjustments, such as changing the address selection jumper or solder bridge. This feature makes it possible to connect both DAC modules to the same I2C bus without address conflicts. As a result, the microcontroller can control both analog references using only the SDA and SCL communication lines, reducing the number of required pins while maintaining a simple and expandable digital interface architecture.

XI-A3 User Interface and Communication

The user interface is composed of a 20x4 display, a 3x4 matrix keypad, two dedicated push buttons, and three status indicator LEDs. The matrix keypad allows the user to enter specific voltage and current setpoints directly, providing a faster and more precise adjustment

method compared to interfaces based only on rotary encoders or incremental selection. The 20x4 display presents the configured setpoints as well as the real-time output measurements, allowing the user to monitor both the desired operating conditions and the actual behavior of the power supply. In addition, the two push buttons provide quick access to essential operating functions: one is used to enable or disable the output, while the other allows the user to switch between voltage and current configuration modes. Finally, the three indicator LEDs provide immediate visual feedback regarding the current state of the equipment, showing whether the output is enabled and whether the power supply is operating in constant-voltage (CV) or constant-current (CC) mode. This interface structure improves usability, reduces ambiguity during operation, and supports the educational purpose of the prototype by making the interaction with the power supply clear and intuitive.

XII 3D Mechanical Design Considerations

XII-A Enclosure and Physical Integration

At this stage, no detailed mechanical or 3D design is proposed. The main mechanical constraints, including form factor, enclosure dimensions, and integration requirements, have already been defined in previous sections of this document.

The validation of the mechanical design will be directly driven by the final electronic implementation, particularly the PCB layout, component placement, thermal requirements, and connector positioning. Therefore, the enclosure design is intentionally deferred until the electrical design reaches a sufficient level of maturity to ensure proper fitting, cooling, and usability.

This approach avoids premature mechanical decisions that could constrain the electronic design or require redesign iterations.

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